

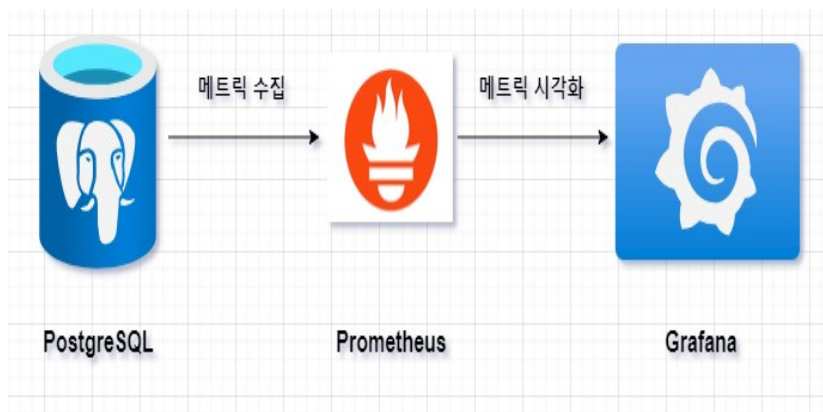
그라파나 구성 가이드

Grafana

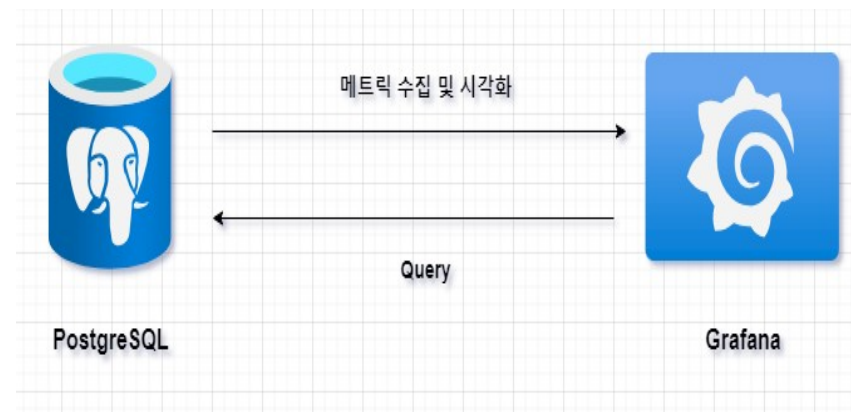
- 오픈소스 메트릭 데이터 시각화 및 메트릭 분석 플랫폼
- 외부 데이터 소스를 정의하고 해당 데이터 소스에 쿼리를 통하여 데이터를 동적으로 가지고 와서 시각화하고 대시보드 구성



- 가장 기본적인 구성으로 [첨부 1] 과 같이 Prometheus 에서 사용할 메트릭 정보를 Agent 를 통하여 PostgreSQL 외부에 노출하고 해당 메트릭 정보를 Prometheus 가 주기적으로 pull 하는 방식으로 메트릭을 수집하며 Prometheus 에서 수집된 메트릭 정보는 Grafana 가 가져가서 시각화하고 대시보드를 구성한다 .
- 본 내용에서는 **데이터 소스 (Database) 의 메트릭 정보를 Grafana 가 어떻게 활용하여 대시보드를 구성하는지에 대한 내용**이므로 [첨부 2] 와 같이 복잡한 설정 작업 없이 **SQL Query** 및 **Grafana Plug-in** 기능 등을 통하여 구성한다 .



[첨부
1]



[첨부
2]

Grafana 대시보드

- 최종 목표 구성 화면



Data Source Summary

- 본 자료에서 다루는 Data Source 는 PostgreSQL DB 임

(Data Source 가 Cloud, DB 인지는 중요하지 않음 -> 어떤 방식으로든 메트릭 정보를 수집 및 가공하여 그라파나 내에서 효과적으로 데이터 시각화 대시보드를 구성하면 되는 거임)

: 본 자료에서 설명되는 메트릭 정보들은 Data Source 로부터 게더링된 쿼리 데이터임

: 본 자료와 같이 다이렉트 쿼리 방식으로 구성할 경우 Data Source 내 OS / DB 등 각 메트릭 데이터에 대하여 별도 테이블 /Function 을 구성하는게 좋음

(각 DB 카탈로그 내 메트릭 정보를 제공하는 오브젝트가 없거나 혹은 원하는 데이터 형식이 다르기에 사용자가 별도 정의 필요)

: 그라파나는 다양한 Data Source 와 연동할 수 있으며 중간에 Prometheus 를 사용하여 구성할 수도 있음

: 본 자료에서 언급된 내용 외 Override, Alert , PlayList 등 많은 기능들이 존재함

: 여러 Plugin 을 통하여 기본 그라파나 디자인 외 프론트엔드 작업이 가능함

: 그라파나 레퍼런스 자료들이 많으므로 꼭 한번 읽어보길 바람 (특히 공식문서)

Grafana 구성 - 1

- Grafana 다운로드 페이지 접속 및 OS/배포판에 적합한 옵션 선택

: <https://grafana.com/grafana/download>

grafana.com/grafana/download

Grafana

Overview Grafana project Grafana Alerting **Download**

10k metrics, 50GB logs, 50GB traces, & more.

Version: 11.0.0
Edition: Enterprise
License: Grafana Labs License
Release Date: May 14, 2024
Release Info: What's New In Grafana 11.0.0

Linux Windows Mac Docker Linux on ARM64

Ubuntu and Debian (64 Bit) SHA256: 42c576ba4e5fad6ddbc68301a983f28e459256498eb655b4229a5ce660d1f70a

```
sudo apt-get install -y adduser libfontconfig1 musl
wget https://dl.grafana.com/enterprise/release/grafana-enterprise_11.0.0_amd64.deb
sudo dpkg -i grafana-enterprise_11.0.0_amd64.deb
```

Read the Ubuntu / Debian [installation guide](#) for more information. We also provide an [APT](#) package repository.

Standalone Linux Binaries (64 Bit) SHA256: 6b80cd453bf834d17106817e23b481b9c8c303ce75670e03e279e2231b1a12ba

```
wget https://dl.grafana.com/enterprise/release/grafana-enterprise-11.0.0.linux-amd64.tar.gz
tar -zxvf grafana-enterprise-11.0.0.linux-amd64.tar.gz
```

Red Hat, CentOS, RHEL, and Fedora (64 Bit) SHA256: 26964ac20e6dd0c3c89850e6e3e3e8200200e6b18f0a1523c5f79b7079d66b39

```
sudo yum install -y https://dl.grafana.com/enterprise/release/grafana-enterprise-11.0.0-1.x86_64.rpm
```

Grafana 구성 - 2

- 패키지 매니저를 통하여 그라파나 설치

```
grafana-enterprise-11.0.0-1.x86_64.rpm 16 MB/s | 114 MB 00:07
경고 : (null): notification message: recovered 395 frames from WAL file /var/lib/dnf/history.sqlite-wal
중속성이 해결되었습니다.

=====
  패키지                                구조                                버전                                저장소                                크기
=====
설치 중 :
grafana-enterprise                      x86_64                            11.0.0-1                            @commandline                            114 M
=====
연결 요약
=====
설치 1 패키지

전체 크기 : 114 M
설치된 크기 : 424 M
패키지 내려받기 중 :
연결 확인 실행 중
연결 확인에 성공했습니다.
연결 시험 실행 중
연결 시험에 성공했습니다.
연결 실행 중
준비 중 :
설치 중 : grafana-enterprise-11.0.0-1.x86_64 1/1
구현 중 : grafana-enterprise-11.0.0-1.x86_64 1/1
### NOT starting on installation, please execute the following statements to configure grafana to start automatically using systemd
sudo /bin/systemctl daemon-reload
sudo /bin/systemctl enable grafana-server.service
### You can start grafana-server by executing
sudo /bin/systemctl start grafana-server.service

POSTTRANS: Running script

확인 중 : grafana-enterprise-11.0.0-1.x86_64 1/1

설치되었습니다 :
grafana-enterprise-11.0.0-1.x86_64

완료되었습니다 !
[root@edb01 ~]#
```

Grafana 구성 - 3

- 그라파나 서비스 실행 및 상태 확인

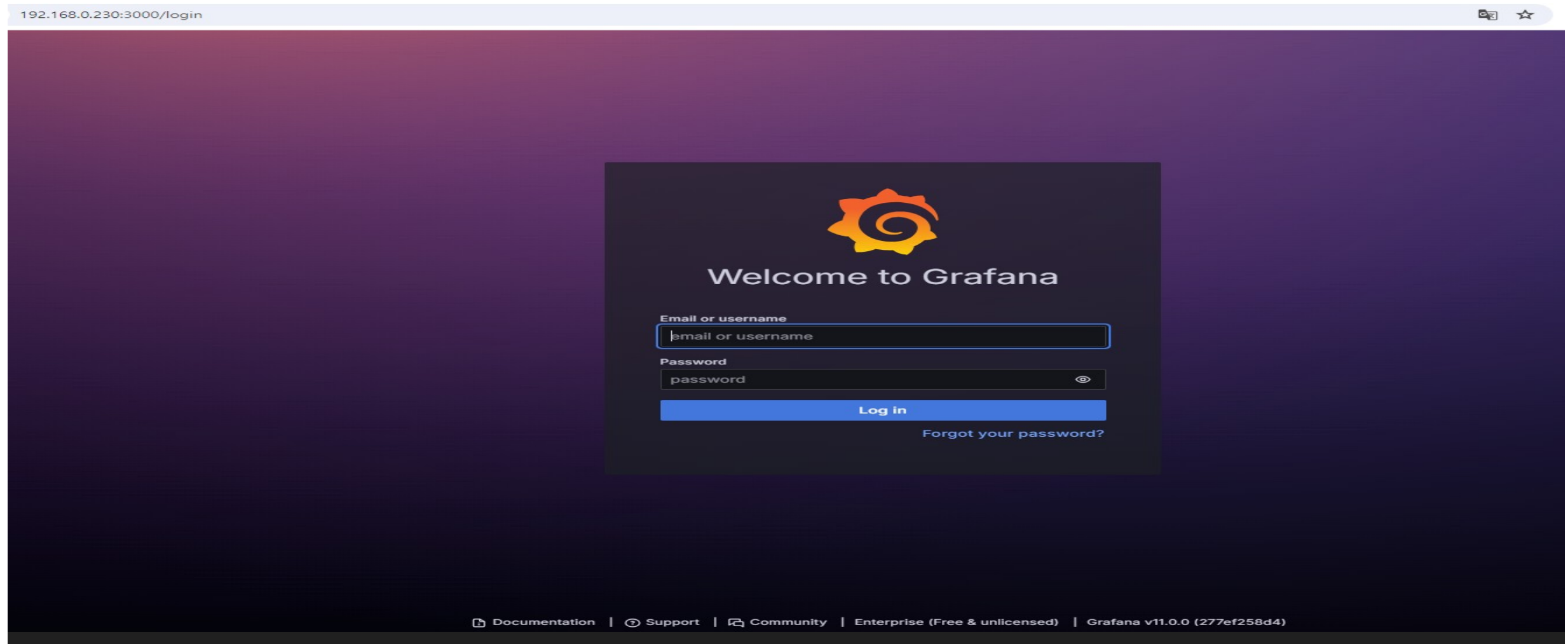
```
[root@edb01 ~]# systemctl start grafana-server.service
[root@edb01 ~]#
[root@edb01 ~]#
[root@edb01 ~]# systemctl status grafana-server.service
● grafana-server.service - Grafana instance
   Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; disabled; preset: disabled)
   Active: active (running) since Tue 2024-06-11 17:20:39 KST; 4s ago
     Docs: http://docs.grafana.org
  Main PID: 224812 (grafana)
    Tasks: 36 (limit: 408473)
   Memory: 69.5M
      CPU: 2.030s
   CGroup: /system.slice/grafana-server.service
           └─224812 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini --pidfile=/var/run/grafana/grafana-server.pid --packaging=rpm cfg:default.paths.logs=/var/log/grafana

6월 11 17:20:39 edb01 grafana[224812]: logger=ngalert.state.manager t=2024-06-11T17:20:39.578453926+09:00 level=info msg="State cache has been initialized" states=0 duration=66.384569ms
6월 11 17:20:39 edb01 grafana[224812]: logger=provisioning.dashboard t=2024-06-11T17:20:39.578505456+09:00 level=info msg="starting to provision dashboards"
6월 11 17:20:39 edb01 grafana[224812]: logger=ngalert.scheduler t=2024-06-11T17:20:39.578530028+09:00 level=info msg="Starting scheduler" tickInterval=10s maxAttempts=1
6월 11 17:20:39 edb01 grafana[224812]: logger=provisioning.dashboard t=2024-06-11T17:20:39.578545217+09:00 level=info msg="finished to provision dashboards"
6월 11 17:20:39 edb01 grafana[224812]: logger=ticker t=2024-06-11T17:20:39.578587187+09:00 level=info msg="starting first_tick=2024-06-11T17:20:40+09:00"
6월 11 17:20:39 edb01 grafana[224812]: logger=grafana.update.checker t=2024-06-11T17:20:39.779847661+09:00 level=info msg="Update check succeeded" duration=267.64871ms
6월 11 17:20:39 edb01 grafana[224812]: logger=grafana-apiserver t=2024-06-11T17:20:39.796593811+09:00 level=info msg="Adding GroupVersion playlist.grafana.app v0alpha1 to ResourceManager"
6월 11 17:20:39 edb01 grafana[224812]: logger=grafana-apiserver t=2024-06-11T17:20:39.797119414+09:00 level=info msg="Adding GroupVersion featuretoggle.grafana.app v0alpha1 to ResourceManager"
6월 11 17:20:39 edb01 grafana[224812]: logger=plugin.angular detectorsprovider.dynamic t=2024-06-11T17:20:39.805438079+09:00 level=info msg="Patterns update finished" duration=225.644139ms
6월 11 17:20:39 edb01 grafana[224812]: logger=plugins.update.checker t=2024-06-11T17:20:39.810794819+09:00 level=info msg="Update check succeeded" duration=298.681246ms
[root@edb01 ~]#
```

Grafana 구성 - 4

- 웹 브라우저 내 URL 접속

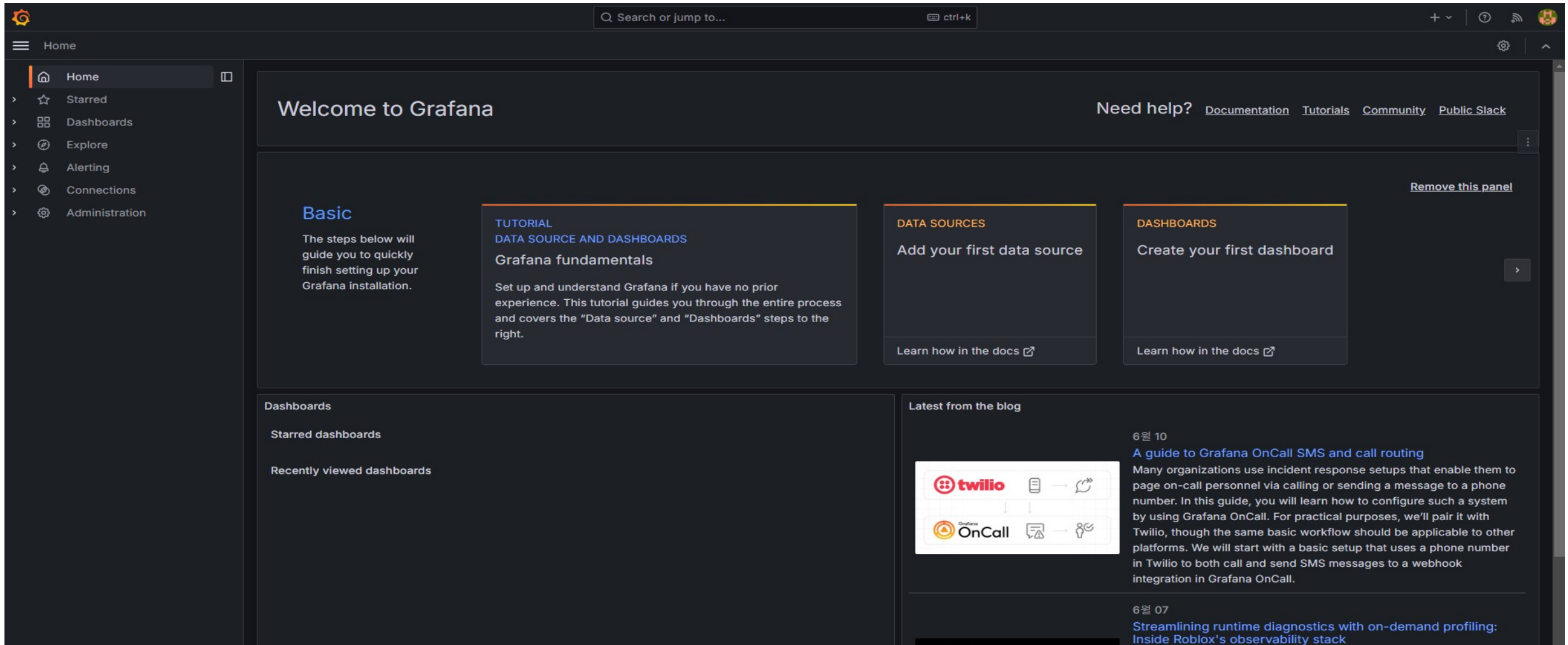
: <http://설치서버IP:3000>



Grafana 구성 - 5

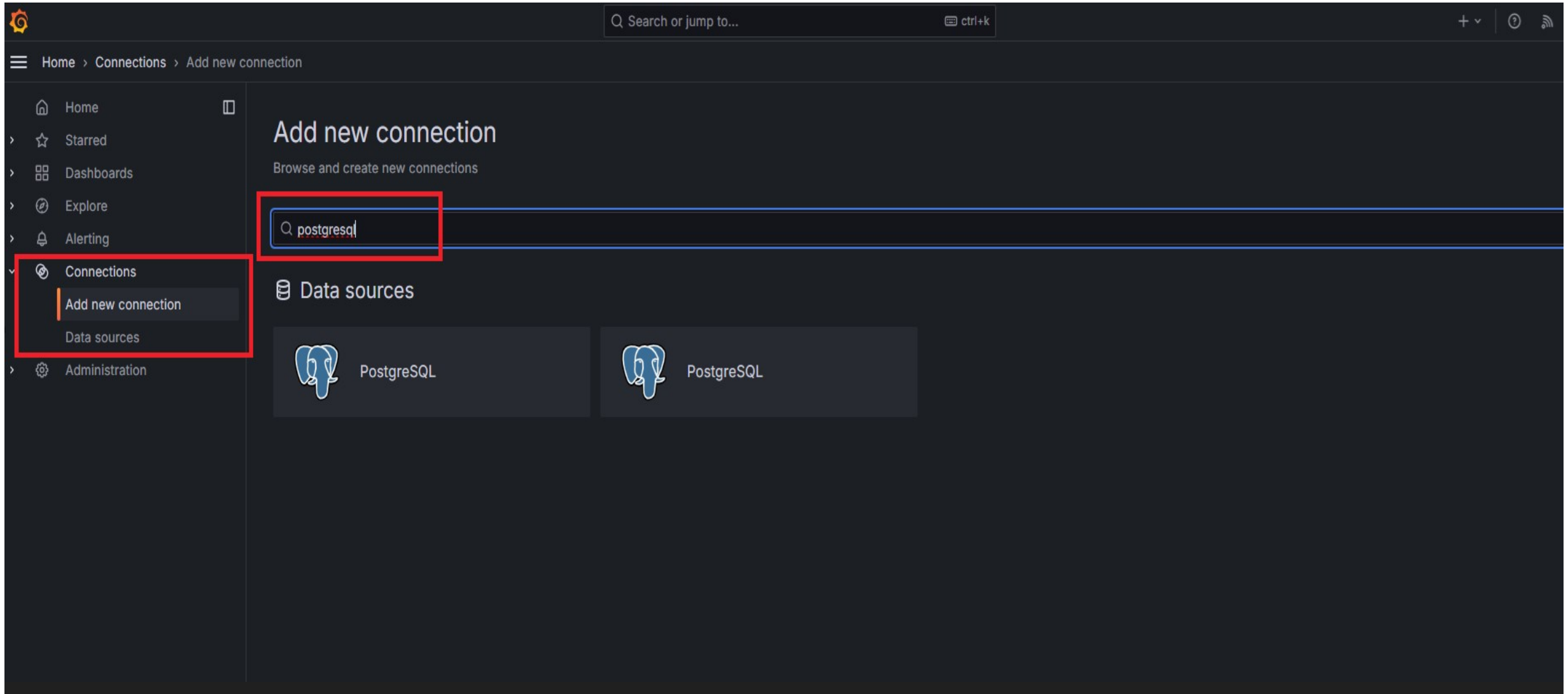
- Default 접속 정보 입력 후 로그인 (이후 Password 변경 화면은 Skip)

: ID/PW -> admin / admin



Grafana 구성 - 6

- 좌측 Connections 내 Add new connection 선택 -> postgresql 입력



Grafana 구성 - 7

- PostgreSQL Data Source 추가

The screenshot shows the Grafana interface for configuring a PostgreSQL data source. The left sidebar contains the following menu items: Home, Starred, Dashboards, Explore, Alerting, Connections, Add new connection, PostgreSQL Data Source 추가, Data sources (highlighted), and Administration. The main content area is titled 'PostgreSQL' and includes the text 'Data source for PostgreSQL and compatible databases' and a warning: 'This plugin is not published to grafana.com/plugins and can't be managed via the catalog.' Below this is an 'Overview' tab. In the top right corner, there is a 'From' dropdown set to 'Grafana Labs', a 'Signature' dropdown set to 'Core', and a blue button labeled 'Add new data source' which is highlighted with a red dashed box.

Grafana PostgreSQL Data Source - Native Plugin

Grafana ships with a built-in PostgreSQL data source plugin that allows you to query and visualize data from a PostgreSQL compatible database.

Adding the data source

1. Open the side menu by clicking the Grafana icon in the top header.
2. In the side menu under the Dashboards link you should find a link named Data Sources.
3. Click the + Add data source button in the top header.
4. Select PostgreSQL from the Type dropdown.

<http://docs.grafana.org/features/datasources/postgres/>

Grafana 구성 - 8

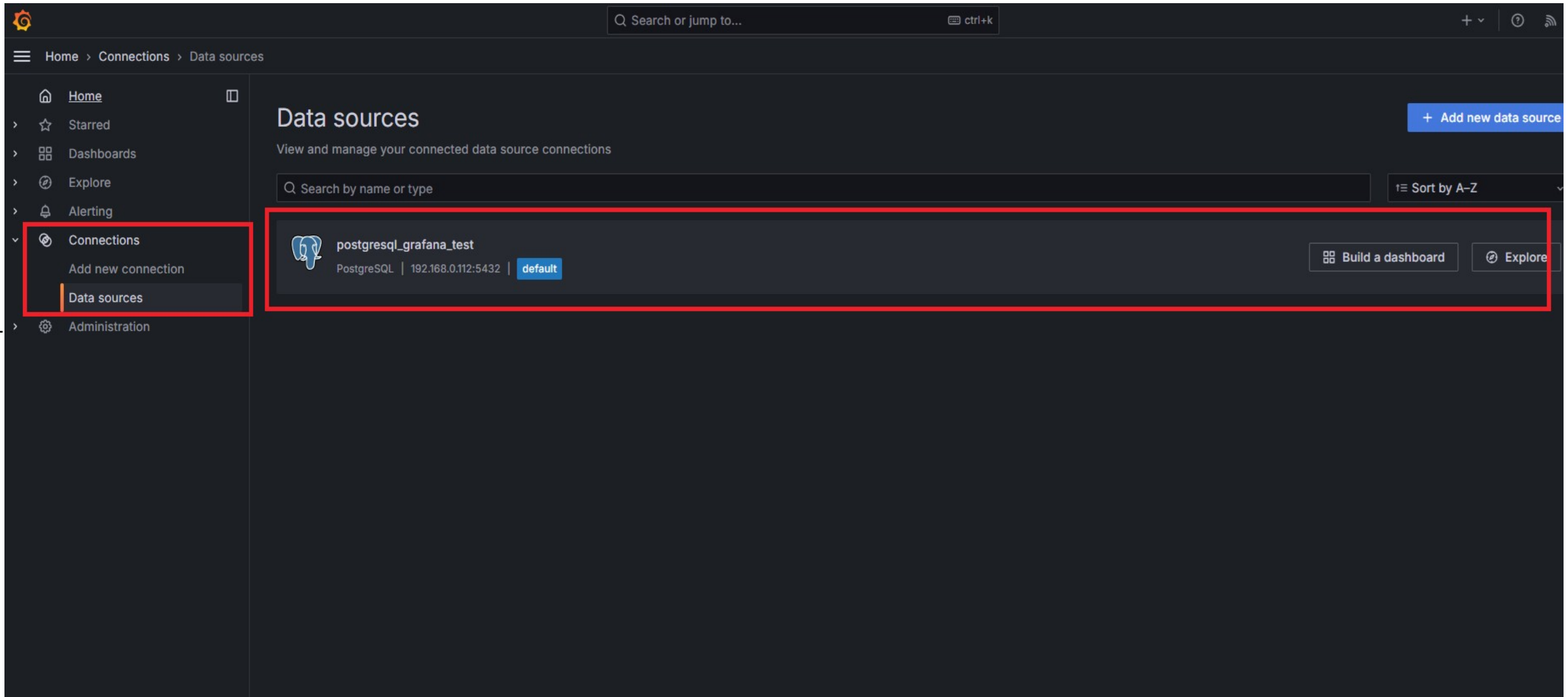
- 사전 구성한 PostgreSQL DB 서버 정보 입력 -> 하단 Save&Test 선택 -> “Database Connection OK” 메시지 출력 시 Connection 정상

The screenshot shows the Grafana Settings page for a PostgreSQL data source. The left sidebar contains navigation links: Starred, Dashboards, Explore, Alerting, Connections, Add new connection, Data sources (highlighted), and Administration. The main content area is titled 'Settings' and contains the following sections:

- Name:** A text input field containing 'postgresql_grafana_test'. A red box highlights this field.
- Default:** A toggle switch currently turned off.
- Instructions:** A paragraph stating: 'Before you can use the Postgres data source, you must configure it below or in the config file. For detailed instructions, [view the documentation](#). Fields marked with * are required'.
- User Permissions:** A section with a dropdown arrow and the title 'User Permissions'. It contains text explaining that the database user should only be granted SELECT permissions and that Grafana does not validate queries. It provides an example of unsafe queries: 'DELETE FROM user;' and 'DROP TABLE user;'. A red box highlights the 'Host URL' and 'Database name' fields.
- Connection:** A section containing two text input fields: 'Host URL *' with the value '192.168.0.112:5432' and 'Database name *' with the value 'gpperfmon'. A red box highlights these two fields.
- Authentication:** A section containing two text input fields: 'Username *' with the value 'gpmon' and 'Password *' with the value 'configured'. There is a 'Reset' button next to the password field. A red box highlights the 'Username' and 'Password' fields.
- TLS/SSL Mode:** A dropdown menu currently set to 'disable'.

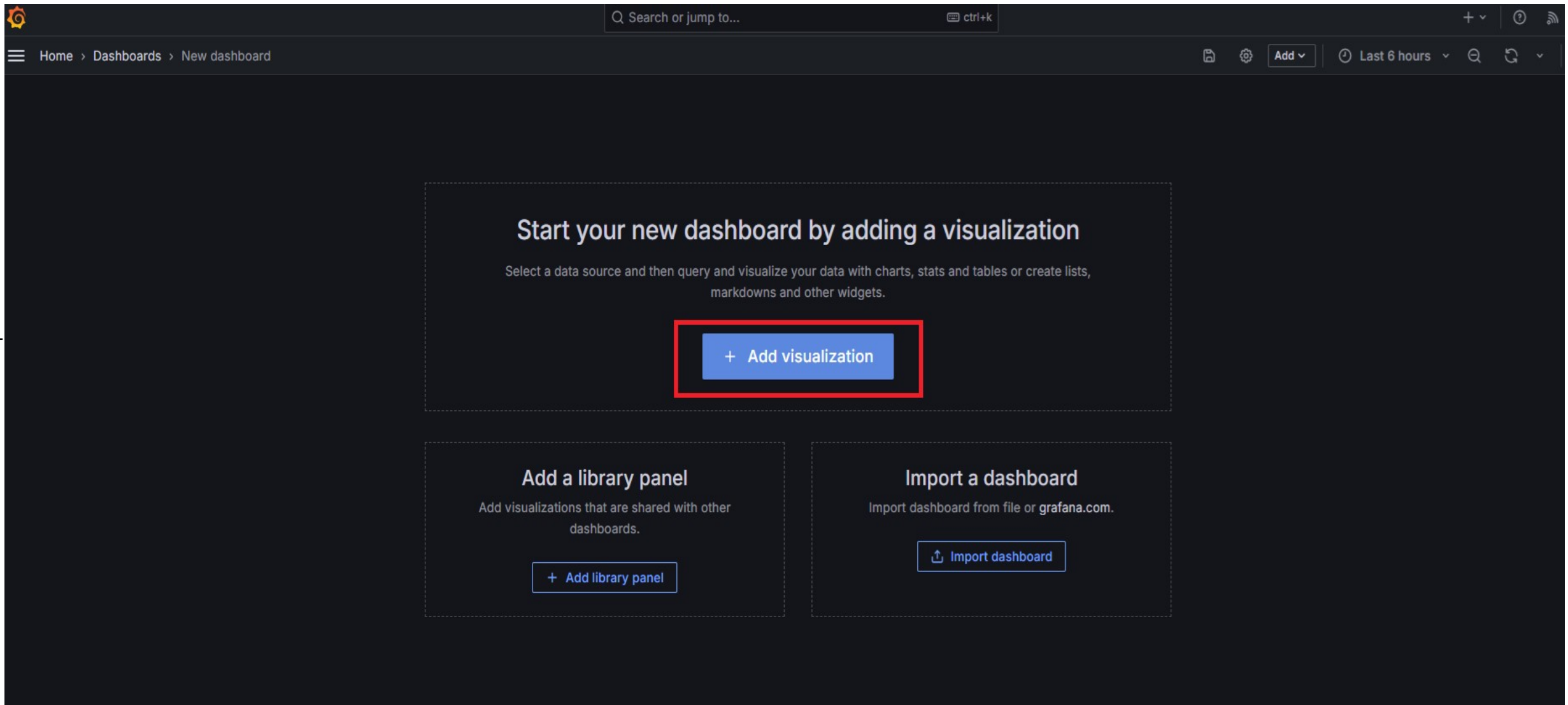
Grafana 구성 - 9

- Data Source 상태 확인 -> “Build a dashboard” 선택



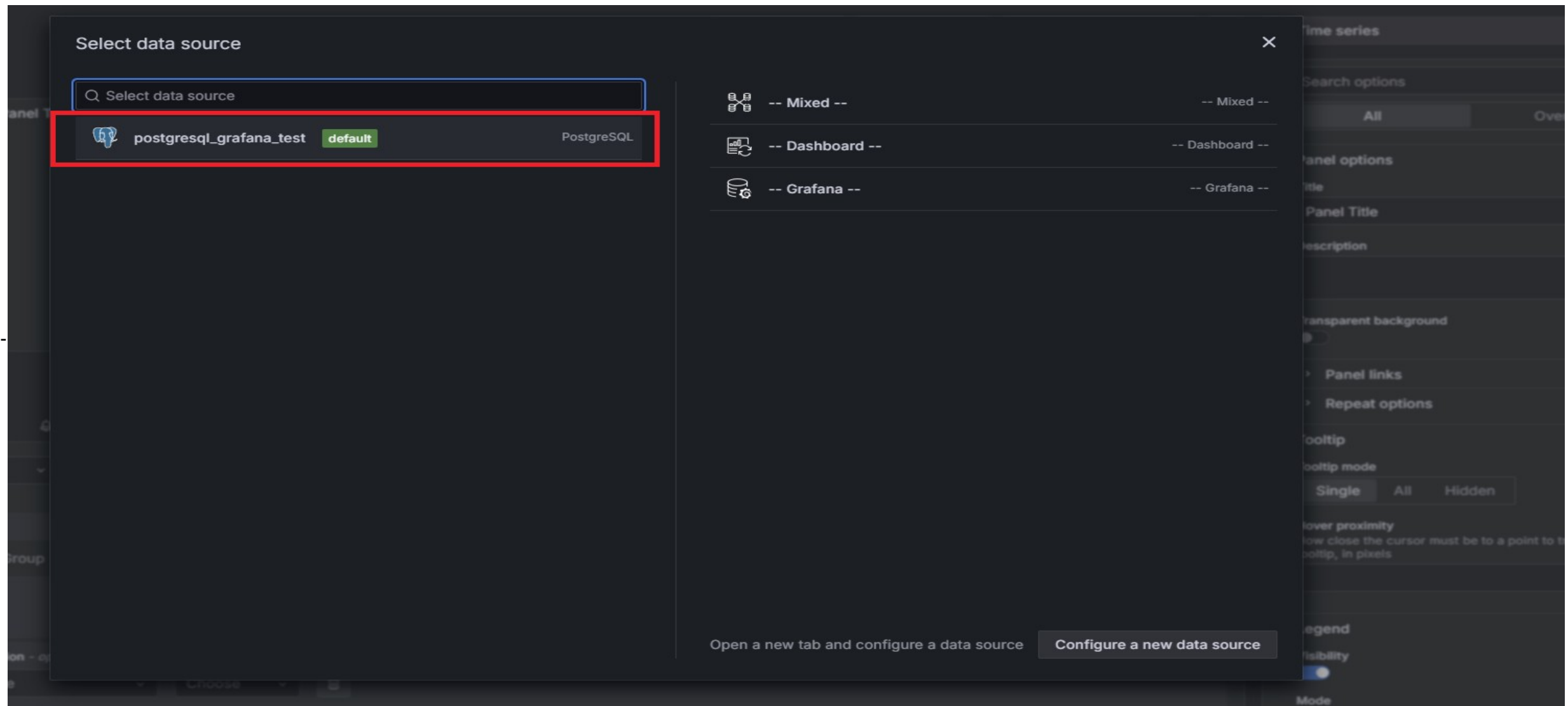
Grafana 구성 - 10

- “Add visualization” 선택



Grafana 구성 - 11

- 기존 구성한 Data Source 선택



Grafana 구성 - 12

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“정상“ 또는 “비정상“)
- 우측 선택 박스를 클릭하여 “Table” visualization 선택 -> Apply 선택

The screenshot shows the Grafana dashboard configuration interface. The top navigation bar includes a search bar and a 'ctrl+k' shortcut. The breadcrumb trail indicates the current location: Home > Dashboards > New dashboard > Edit panel. The main panel area displays a table visualization with the title 'Panel Title' and a single row containing the text '정상'. The right sidebar contains the configuration options for the table visualization, including 'Table view' toggle, 'Fill' and 'Actual' buttons, and a 'Table' dropdown menu. The bottom section shows the query editor with the 'Data source' set to 'postgresL_grafana_test' and the 'Query options' section. The query text is: `SELECT CASE WHEN COUNT(*) = 0 THEN '정상' ELSE '비정상' END " " FROM gpstate;`. The 'Code' button in the bottom right corner of the query editor is highlighted.

Panel Title

정상

Data source: postgresL_grafana_test

Query options: MD = auto = 914 Interval = 1m

Format: Table

1 `SELECT CASE WHEN COUNT(*) = 0 THEN '정상' ELSE '비정상' END " " FROM gpstate;`

Table view: ☐ Table view ☒ Fill Actual Last 5 minutes

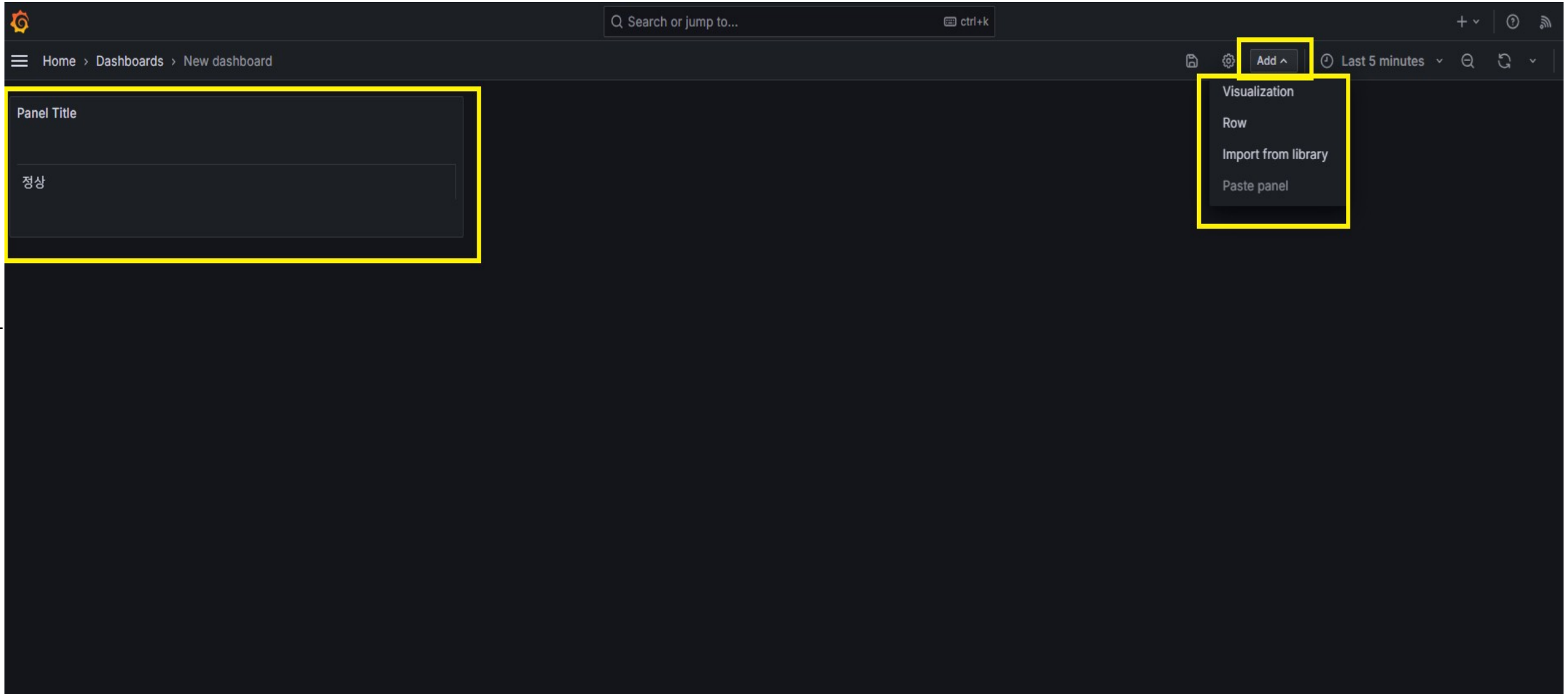
Table visualization options:

- Search options: All Overrides
- Panel options: Title (Panel Title), Description, Transparent background (off)
- Panel links
- Repeat options
- Table: Show table header (on), Cell height (Small Medium Large), Enable pagination (off), Minimum column width

Buttons: Discard Save Apply Run query Builder Code

Grafana 구성 - 13

- 완성된 Panel 확인 -> 드래그를 통하여 패널 이동 및 크기 조절 가능 -> Panel 추가를 위하여 우측 “Add” 박스 클릭 및 Visualization 선택



Grafana 구성 - 14

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“Sync” 또는 “Not Sync”)

The screenshot shows the Grafana 'Edit panel' interface. The top navigation bar includes 'Home > Dashboards > New dashboard > Edit panel'. The right sidebar contains buttons for 'Discard', 'Save', and 'Apply' (highlighted). The main panel area shows a 'Table' view with a 'Panel Title' and a 'Sync' button (highlighted). The bottom section shows the 'Query' editor with a 'Data source' dropdown set to 'postgresql_grafana_test' (highlighted). The 'Query options' section shows 'MD = auto = 914' and 'Interval = 1m'. The 'Code' button in the query editor is highlighted. The SQL query is: `SELECT CASE WHEN VALUE ILIKE '%Active%' THEN 'Sync' ELSE 'Not Sync' END AS \" \" FROM gpstate_summary WHERE descr ilike 'Master instance%%';`

Grafana 구성 - 15

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“DATABASE 이름“)

The image shows the Grafana web interface for editing a dashboard panel. The top navigation bar includes a search bar and buttons for 'Discard', 'Save', and 'Apply'. The breadcrumb trail indicates the current location: 'Home > Dashboards > 20240611 > Edit panel'. The panel being edited is titled 'gpperfmon'. The panel's display mode is set to 'Table', and the data source is 'postgresql_grafana_test'. The query editor at the bottom shows a SQL query: `SELECT CURRENT_DATABASE() AS " ";`. The right sidebar contains various configuration options for the panel, including 'Panel options', 'Table' settings, and 'Query options'.

Panel Title

gpperfmon

Table view ☐ Fill Actual Last 5 minutes ☐ ☐ ☐ Table

Search options

All Overrides

Panel options

Title

Panel Title

Description

Transparent background ☐

Panel links

Repeat options

Table

Show table header ☒

Cell height

Small Medium Large

Enable pagination ☐

Minimum column width

The minimum width for column auto resizing

Query 1 Transform data 0

Data source postgresql_grafana_test

Query options MD = auto = 914 Interval = 1m

Query inspector

Format: Table

Run query Builder Code

SELECT CURRENT_DATABASE() AS " ";

Grafana 구성 - 16

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“DB 버전“)

The screenshot shows the Grafana interface for editing a panel. The panel title is "Panel Title" and it displays the value "6.26.1". The right sidebar shows the "Table" view selected. The bottom query editor shows a SQL query: `SELECT Version[1] as " " FROM (SELECT regexp_matches(VERSION(), '.*Greenplum Database(.*) build commit.*') AS Version) A;`. The query is highlighted in yellow.

Search or jump to... ctrl+k

Home > Dashboards > 20240611 > Edit panel

Discard Save **Apply**

Table view ☐ Fill Actual Last 5 minutes ☐ ☐ ☐

Table

Search options

All Overrides

Panel options

Title

Panel Title

Description

Transparent background ☐

Panel links

Repeat options

Table

Show table header ☒

Cell height

Small Medium Large

Enable pagination ☐

Minimum column width

The minimum width for column auto resizing

Query 1 Transform data 0

Data source postgresql_grafana_test

Query options MD = auto = 914 Interval = 1m

Query inspector

Format: Table

Run query Builder Code

```
1 SELECT Version[1] as " " FROM (SELECT regexp_matches(VERSION(), '.*Greenplum Database(.*) build commit.*') AS Version) A;
```

Grafana 구성 - 17

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“현재 세션 개수“)

The screenshot shows the Grafana interface with a dashboard titled '20240611'. The main panel, titled 'Panel Title', displays a large green number '12' on a dark background, which is highlighted with a yellow rectangle. The top right of the dashboard has buttons for 'Discard', 'Save', and 'Apply', with 'Apply' highlighted in yellow. The right sidebar contains the 'Stat' panel configuration, also highlighted with a yellow rectangle. The bottom section shows the 'Query' editor with the data source set to 'postgresql_grafana_test' (highlighted with a yellow rectangle) and the query 'SELECT COUNT(sess_id)-1 AS " " FROM pg_catalog.pg_stat_activity' (highlighted with a yellow rectangle). The query editor also shows 'Query options' (MD = auto = 914, Interval = 1m) and a 'Run query' button.

Grafana 구성 - 18

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“현재 인스턴스 상태값”)

The screenshot displays the Grafana interface with a SQL query editor at the bottom and a table visualization in the center. The table shows the results of a query about segment status.

Table View:

Segment Status	Count
Up	16
Down	0

SQL Query:

```
1 SELECT CASE status WHEN 'u' THEN 'Up' WHEN 'd' THEN 'Down' END AS "Segment Status", MAX(count) AS "Count"
2 FROM (
3   SELECT status, COUNT(*)
4   FROM pg_catalog.gp_segment_configuration
5   WHERE content <> -1
6   GROUP BY status
7   UNION
8   SELECT 'd', 0
9   UNION
10  SELECT 'u', 0) test
11 GROUP BY status
12 ORDER BY status desc;
```

Right Panel Options:

- Table:** Search options, All, Overrides
- Panel options:** Title (Panel Title), Description
- Transparent background:** Toggle off
- Panel links:** >
- Repeat options:** >
- Table:** Show table header (Toggle on), Cell height (Small, Medium, Large), Enable pagination (Toggle off), Minimum column width (150), Column width

Grafana 구성 - 19

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“현재 인스턴스 Sync 상태”)

The screenshot displays the Grafana interface with a table visualization and the query editor. The table visualization is titled "Panel Title" and shows the following data:

Segment Role	Count
Synchronized	16
Not In Sync	0

The query editor at the bottom shows the following SQL query:

```
1 SELECT CASE WHEN "role" = preferred_role THEN 'Synchronized' WHEN "role" != preferred_role THEN 'Not In Sync' END AS "Segment Role", SUM(count) AS "Count"
2 FROM (
3   SELECT "role", preferred_role, COUNT(*)
4   FROM pg_catalog.gp_segment_configuration
5   WHERE content <> -1
6   GROUP BY "role", preferred_role
7   UNION
8   SELECT 'p', 'p', 0
9   UNION
```

The right sidebar shows the "Table" panel options, including "Show table header" (checked), "Cell height" (Small, Medium, Large), "Enable pagination" (unchecked), "Minimum column width" (150), and "Column width" (auto).

Grafana 구성 - 20

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“DB 스키마 개수“)

The screenshot displays the Grafana web interface in a dark theme. At the top, the breadcrumb navigation shows 'Home > Dashboards > 20240611 > Edit panel'. In the top right corner, there are buttons for 'Discard', 'Save', and 'Apply'. Below the navigation, a toolbar contains 'Table view' (with a toggle), 'Fill', 'Actual', a time range selector set to 'Last 5 minutes', a search icon, and a refresh icon. The main panel area shows a configuration for a table visualization. A yellow box highlights the 'Panel Title' field, which contains the text 'count' and '4 EA'. To the right, a sidebar contains the 'Table' configuration panel, which is also highlighted with a yellow box. It includes a search bar for options, tabs for 'All' and 'Overrides', and sections for 'Panel options' (Title, Description), 'Transparent background' (toggle), 'Panel links', 'Repeat options', and 'Table' settings (Show table header, Cell height, Enable pagination, Minimum column width). At the bottom, a 'Query' editor is visible, showing a SQL query for counting namespaces in a PostgreSQL database. The query is:

```
1 SELECT count(*) || ' ' || 'EA' as count
2 FROM pg_catalog.pg_namespace n
3 WHERE n.nspname !~ '^pg_' AND n.nspname <> 'information_schema'
4 ORDER BY 1;
```

 The query editor has a 'Format' dropdown set to 'Table', a 'Run query' button, and 'Builder' and 'Code' tabs. A yellow dashed box highlights the entire query editor area.

Home > Dashboards > 20240611 > Edit panel

Table view ☐ Fill Actual Last 5 minutes 🔍 ↻

Panel Title

count

4 EA

Search options

All Overrides

Panel options

Title

Panel Title

Description

Transparent background ☐

Panel links

Repeat options

Table

Show table header ☒

Cell height

Small Medium Large

Enable pagination ☐

Minimum column width

The minimum width for column auto resizing

150

Query 1 Transform data 0

Data source postgresql_grafana_test ⓘ

Query options MD = auto = 914 Interval = 1m Query inspector

Format: Table ▼

Run query Builder Code

```
1 SELECT count(*) || ' ' || 'EA' as count
2 FROM pg_catalog.pg_namespace n
3 WHERE n.nspname !~ '^pg_' AND n.nspname <> 'information_schema'
4 ORDER BY 1;
```

Grafana 구성 - 21

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“DB 테이블 개수“)

The screenshot shows the Grafana interface for editing a panel. The top navigation bar includes 'Home > Dashboards > 20240611 > Edit panel'. The panel title is 'Panel Title'. The main display area shows a table with one row containing the value '152 EA'. The right sidebar contains the 'Table' configuration options, including 'Show table header' (checked), 'Cell height' (Small, Medium, Large), 'Enable pagination' (unchecked), and 'Minimum column width' (150). The bottom section shows the query editor with the following SQL query:

```
1 SELECT count(*) || ' ' || 'EA' as " "
2 FROM pg_catalog.pg_class c
3 | LEFT JOIN pg_catalog.pg_namespace n ON n.oid = c.relnamespace
4 WHERE c.relkind IN ('f','p','t')
5 AND c.relstorage IN ('x','f','h','a','c','')
6 | AND n.nspname <> 'pg_catalog'
```

The query is executed using the 'Run query' button. The results are displayed in a table format.

Grafana 구성 - 22

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“DB 서버 Uptime 정보“)

The screenshot shows the Grafana interface for editing a panel. The main panel displays the text "98 days 03:57:28". The sidebar on the right contains configuration options for the panel, including "Table" view, "Search options", "Panel options", and "Table" settings. The bottom of the interface shows the query editor with a SQL query and a "Run query" button.

Panel Title

98 days 03:57:28

Table view ☐ Fill Actual Last 5 minutes

Table

Search options

All Overrides

Panel options

Title

Panel Title

Description

Transparent background

Panel links

Repeat options

Table

Show table header

Cell height

Small Medium Large

Enable pagination

Query 1 Transform data 0

Data source postgresql_grafana_test

Query options MD = auto = 914 Interval = 1m

Query inspector

Format: Table

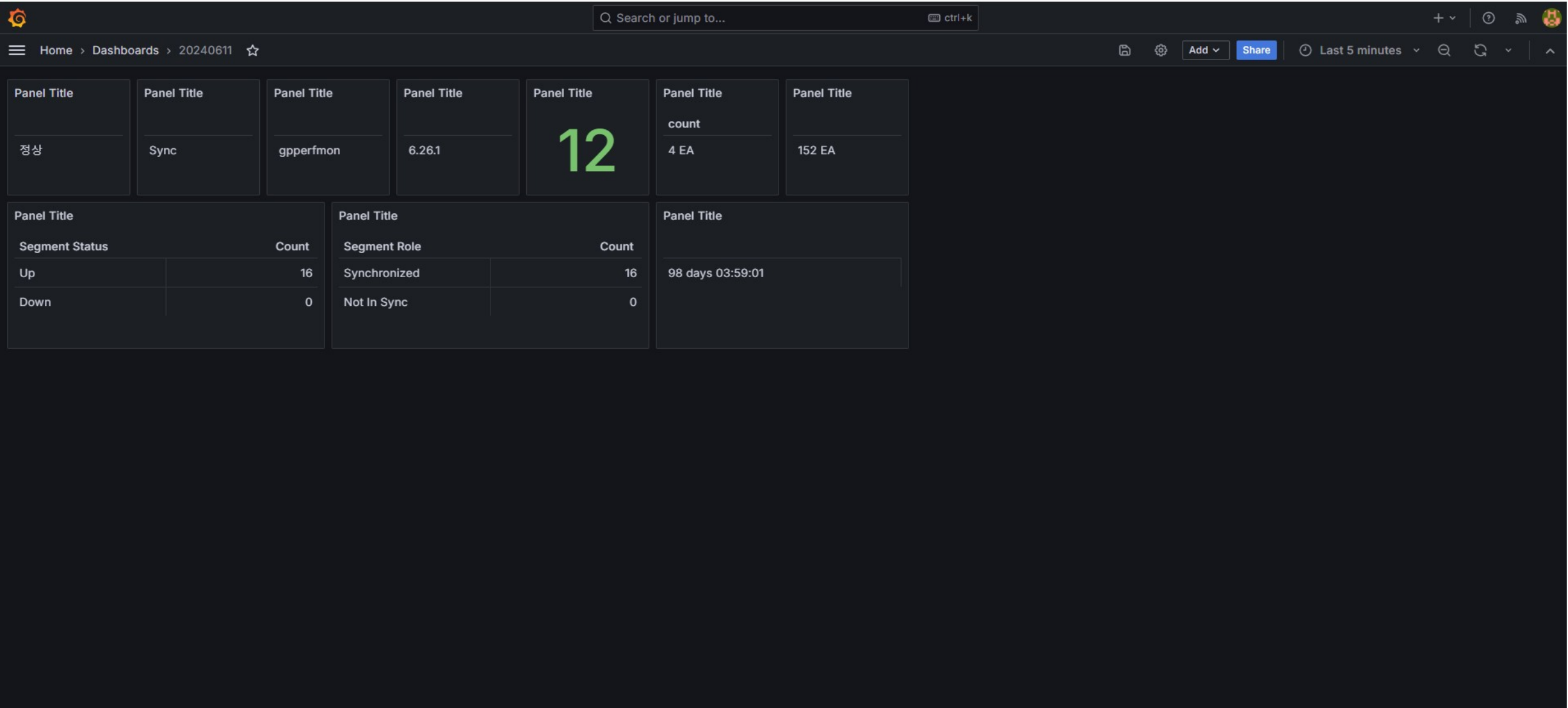
Run query

Builder Code

```
1 SELECT date_trunc('second', current_timestamp - pg_postmaster_start_time()) as " ";
```

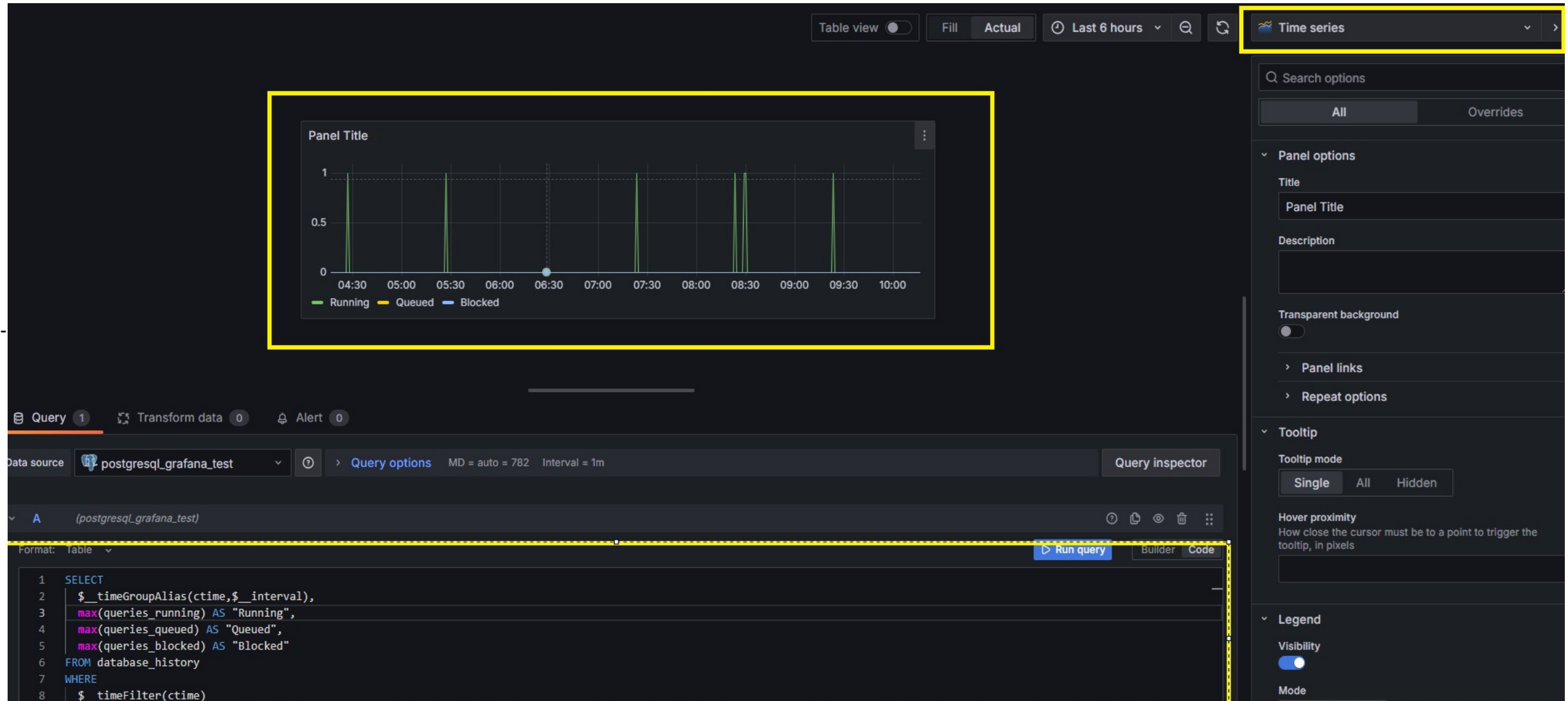
Grafana 구성 - 23

- 구성 화면 확인



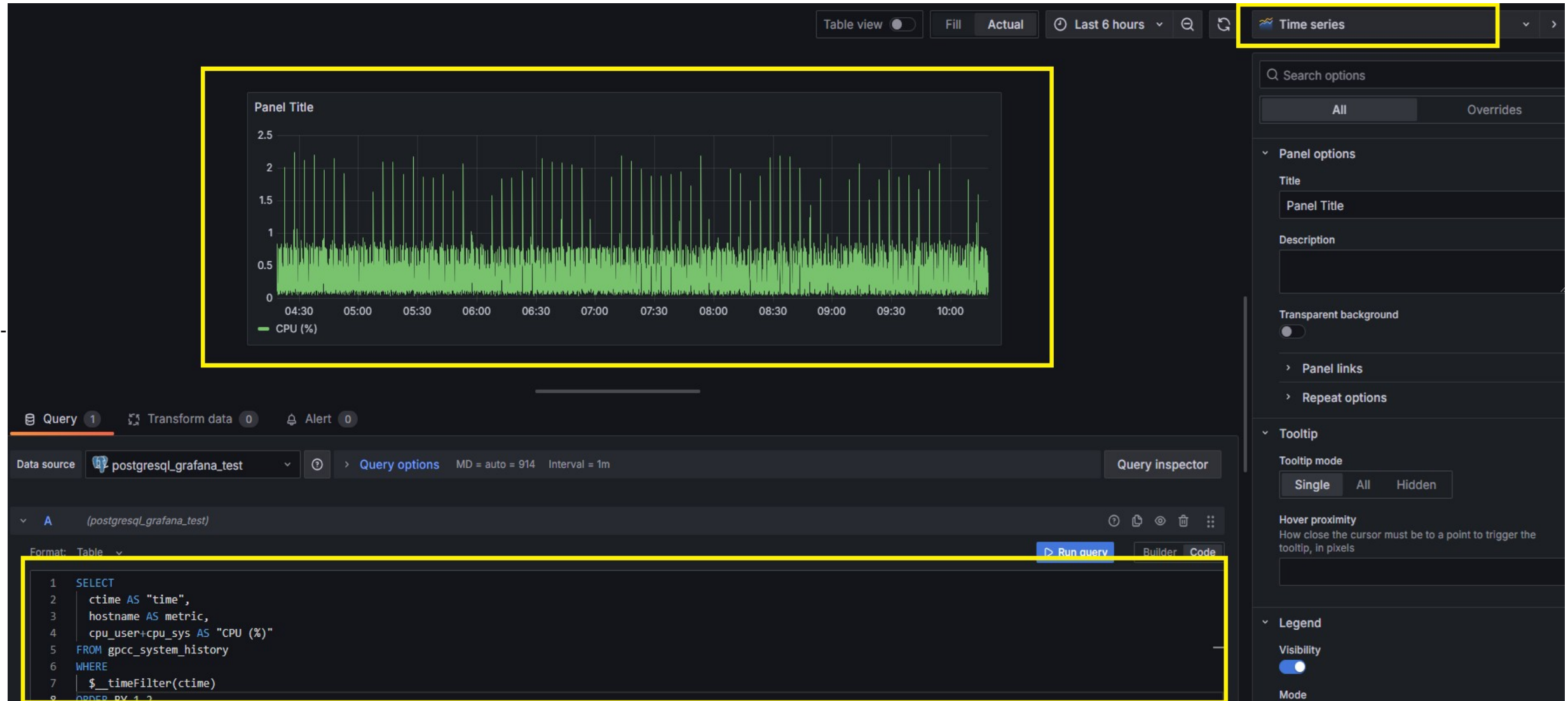
Grafana 구성 - 24

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“대기 / 활성 / 만료 세션 확인“)



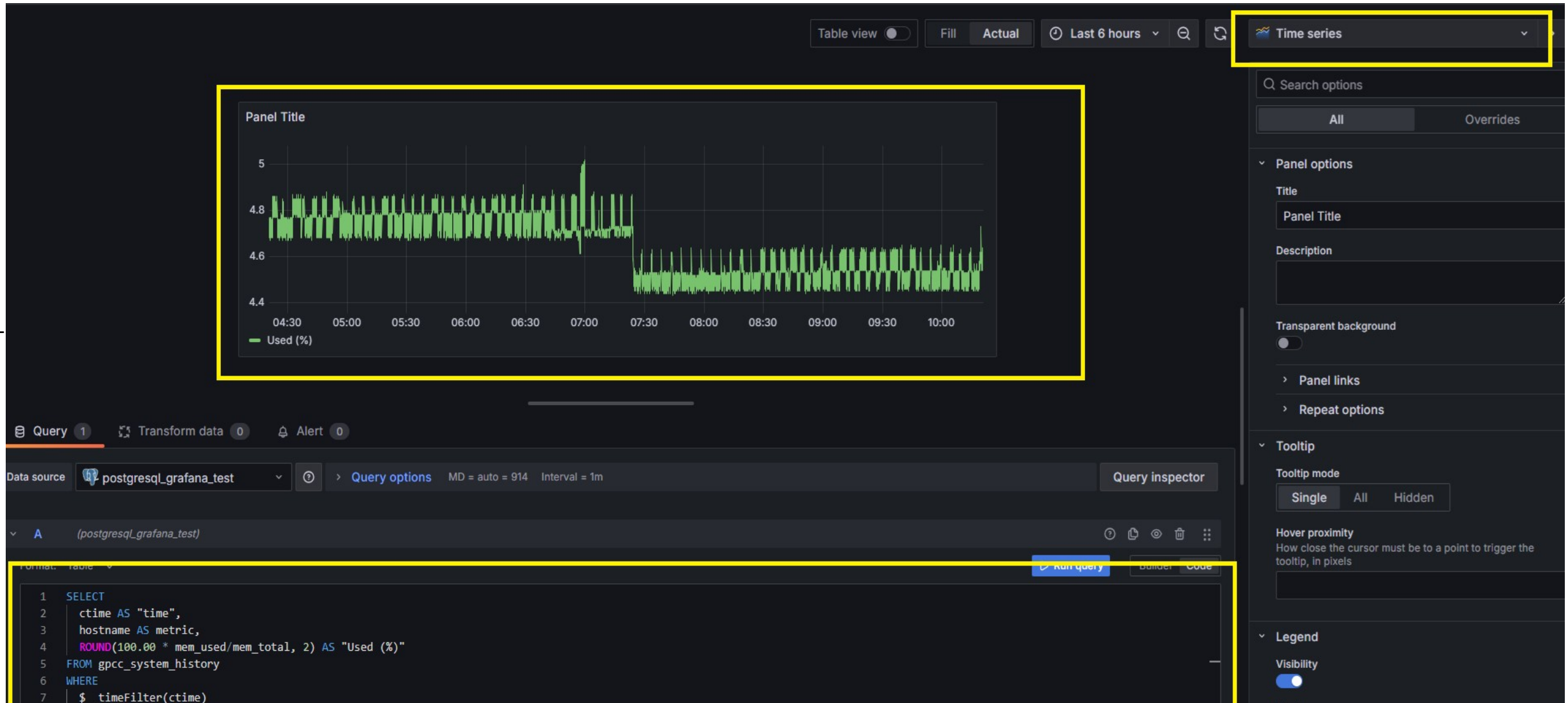
Grafana 구성 - 25

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“시스템 리소스 - CPU“)



Grafana 구성 - 26

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“시스템 리소스 - Memory“)



Grafana 구성 - 27

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 (“시스템 리소스 – DISK I/O”)



Grafana 구성 - 28

- 우측 선택박스 내 “TEXT” 선택 -> 하단 HTML 선택 -> HTML 작성 -> 패널 디스플레이에 코드 결과값이 나오는지 확인

The screenshot displays the Grafana interface with a dark theme. On the left, a panel titled "Panel Title" is highlighted with a yellow border. It contains the text "블로트가 의심되는 테이블이 있어 VACUUM 작업이 필요합니다." and a horizontal line. The right sidebar is also highlighted with a yellow border and contains the configuration for the selected "Text" panel. At the top of the sidebar, the "Text" panel type is selected. Under "Panel options", the "Title" is set to "Panel Title" and the "Description" field is empty. The "Transparent background" toggle is turned off. Under "Text" options, the "Mode" is set to "HTML", which is highlighted with a yellow box. The "Content" field at the bottom is highlighted with a yellow box and contains the following HTML code: `<p style="font-size: 15px; font-weight: bold; border-bottom: 1px solid #688FF4; padding: 0.1em; color: whi`. The top of the interface shows the "Table view" toggle, "Fill" and "Actual" buttons, a time range of "Last 6 hours", and search and refresh icons.

Grafana 구성 - 29

- 우측 선택박스 내 “TEXT” 선택 -> 하단 HTML 선택 -> HTML 작성 -> 패널 디스플레이에 코드 결과값이 나오는지 확인

The screenshot displays the Grafana interface for configuring a panel. The main panel area on the left shows a preview of the panel with the title "Panel Title" and the text "블로트가 의심되는 테이블이 있어 VACUUM FULL 작업이 필요합니다." Below the text is a horizontal line. The right sidebar contains the configuration options for the panel. The "Text" option is selected in the top right corner. Under the "Text" section, the "Mode" is set to "HTML". The "Content" field is visible at the bottom, showing the HTML code: `<p style="font-size: 15px; font-weight: bold; border-bottom: 1px solid #688FF4; padding: 0.1em; color: whi`. The "Panel options" section includes fields for "Title" (set to "Panel Title") and "Description". The "Transparent background" toggle is turned off. The "Panel links" and "Repeat options" sections are collapsed.

Table view ☐ Fill Actual Last 6 hours 🔍 ↺

Text

Search options

Panel options

Title

Panel Title

Description

Transparent background

Panel links

Repeat options

Text

Mode

Markdown HTML Code

Content

`<p style="font-size: 15px; font-weight: bold; border-bottom: 1px solid #688FF4; padding: 0.1em; color: whi`

Panel Title

블로트가 의심되는 테이블이 있어 VACUUM FULL 작업이 필요합니다.

Grafana 구성 - 30

- 우측 선택박스 내 “TEXT” 선택 -> 하단 HTML 선택 -> HTML 작성 -> 패널 디스플레이에 코드 결과값이 나오는지 확인

The screenshot displays the Grafana interface with a dark theme. At the top, there are controls for 'Table view' (disabled), 'Fill' and 'Actual' buttons, a time range of 'Last 6 hours', and search and refresh icons. The main panel area on the left shows a 'Panel Title' and a text block: '통계정보가 정확하지 않는 테이블이 있어 ANALYZE 작업이 필요합니다.' This panel is highlighted with a yellow border. On the right, the configuration sidebar is open. The 'Text' panel type is selected at the top, also highlighted with a yellow border. Below this, the 'Panel options' section includes fields for 'Title' (set to 'Panel Title') and 'Description'. The 'Transparent background' toggle is turned off. The 'Text' section at the bottom shows 'Mode' set to 'HTML', which is highlighted with a yellow box. Below the mode selection, the 'Content' field contains the HTML code: `<p>shadow: 1px 1px 1.2px midnightblue;>통계정보가 정확하지 않는 테이블이 있어 ANALYZE 작업이 필요합니다.</p>`. This content field is highlighted with a yellow dashed border.

Grafana 구성 - 31

- 우측 선택박스 내 “TEXT” 선택 -> 하단 HTML 선택 -> HTML 작성 -> 패널 디스플레이에 코드 결과값이 나오는지 확인

The screenshot displays the Grafana interface with a panel configuration for a text widget. The panel on the left is titled "Panel Title" and contains the text "스큐가 발생된 테이블이 있어 분산키 변경 (재분산) 작업이 필요합니다." (A table with skewed data exists, so a distributed key change (repartitioning) operation is required.). The right sidebar shows the configuration options for the "Text" widget. The "Text" section is expanded, and the "Mode" is set to "HTML". The "Content" field is highlighted with a yellow dashed border and contains the HTML code: `shadow: 1px 1px 1.2px midnightblue;">스큐가 발생된 테이블이 있어 분산키 변경 (재분산) 작업이 필요합니다.</p>`. The "Panel options" section includes fields for "Title" (Panel Title) and "Description". The "Transparent background" toggle is turned off. The "Panel links" and "Repeat options" sections are collapsed.

Table view ☐ Fill Actual Last 6 hours

Text

Search options

Panel options

Title

Panel Title

Description

Transparent background

Panel links

Repeat options

Text

Mode

Markdown HTML Code

Content

shadow: 1px 1px 1.2px midnightblue;">스큐가 발생된 테이블이 있어 분산키 변경 (재분산) 작업이 필요합니다.</p>

Grafana 구성 - 32

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 ("Vacuum 작업 대상 오브젝트 개수 ")

The image shows a Grafana dashboard interface. The main panel displays a large green number "32". The top of the dashboard has a navigation bar with "Table view" (disabled), "Fill", "Actual", "Last 6 hours", and a refresh button. The right sidebar contains configuration options for the panel, including "Panel options" (Title: Panel Title, Description: empty), "Value options" (Show: Calculate, Calculation: Last *), and "Fields" (Numeric Fields). The bottom section shows the query editor with a SQL query: `1 SELECT COUNT(*) as " " FROM gpmetrics.test('vacuum');`. The query editor has a "Run query" button and a "Code" button. The data source is set to "postgresql_grafana_test".

Panel Title

32

Query 1 Transform data 0

Data source postgresql_grafana_test Query options MD = auto = 914 Interval = 1m Query inspector

Format: Table

1 SELECT COUNT(*) as " " FROM gpmetrics.test('vacuum');

Run query Builder Code

12.4 Stat

Search options

All Overrides

Panel options

Title Panel Title

Description

Transparent background

Panel links

Repeat options

Value options

Show Calculate All values

Calculation Choose a reducer function / calculation

Last *

Fields Select the fields that should be included in the panel

Numeric Fields

Grafana 구성 - 33

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 ("Vacuum Full 작업 대상 오브젝트 개수 ")

The image shows a Grafana dashboard with a dark theme. At the top, there are controls for the panel: 'Table view' (disabled), 'Fill' and 'Actual' buttons, a time range of 'Last 6 hours', and search and refresh icons. The main panel, titled 'Stat', displays a large red number '83'. To the right is a sidebar with configuration options: 'Search options', 'All' and 'Overrides' tabs, 'Panel options' (Title: 'Panel Title', Description: empty), 'Transparent background' (disabled), 'Panel links', 'Repeat options', 'Value options' (Show: 'Calculate a single value per column or series or show each row', Calculation: 'Last *', Fields: empty), and 'Calculate' and 'All values' buttons.

At the bottom, the 'Query' tab is active, showing the data source 'postgresql_grafana_test' and query options 'MD = auto = 914' and 'Interval = 1m'. The query editor contains the following SQL query:

```
1 SELECT COUNT(*) as " " FROM gpmetrics.test('vacuum-full');
```

The query editor also includes a 'Run query' button and 'Builder' and 'Code' tabs.

Grafana 구성 - 34

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 ("Vacuum 작성 대상 오브젝트 개수 ")

The screenshot displays the Grafana web interface. At the top, there are controls for 'Table view' (disabled), 'Fill' and 'Actual' buttons, a time range of 'Last 6 hours', and search and refresh icons. The main panel area is highlighted with a yellow border and contains a large green number '1'. Below this, the 'Query' tab is active, showing a data source of 'postgresql_grafana_test' and a query editor with the SQL: `SELECT COUNT(*) as " " FROM gpmetrics.test('analyze');`. The right-hand sidebar is also highlighted with a yellow border and contains configuration options for the panel, including 'Panel options' (Title, Description, Transparent background) and 'Value options' (Show, Calculation, Fields).

Panel Title

1

Query 1 Transform data 0

Data source postgresql_grafana_test Query options MD = auto = 914 Interval = 1m Query inspector

Format: Table

1 SELECT COUNT(*) as " " FROM gpmetrics.test('analyze');

12.4 Stat

Search options

All Overrides

Panel options

Title

Panel Title

Description

Transparent background

Panel links

Repeat options

Value options

Show

Calculate a single value per column or series or show each row

Calculate All values

Calculation

Choose a reducer function / calculation

Last *

Fields

Select the fields that should be included in the panel

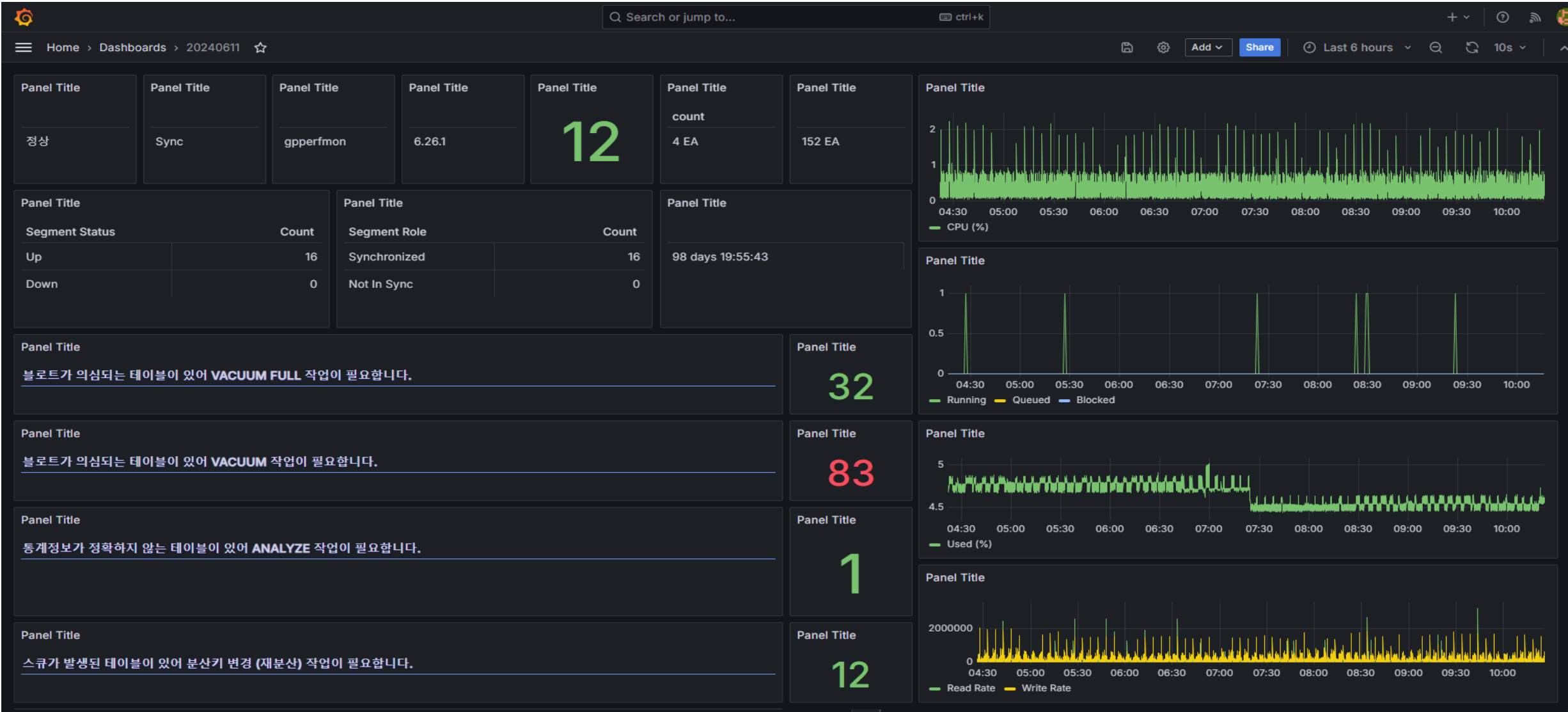
Grafana 구성 - 35

- 하단 텍스트 박스 내 Code 버튼 선택 후 -> 쿼리 입력 -> 중앙 패널 디스플레이에 쿼리 결과값이 나오는지 확인 ("Vacuum 작성 대상 오브젝트 개수 ")

The image shows a Grafana dashboard interface. At the top, there are controls for 'Table view' (disabled), 'Fill' and 'Actual' buttons, a time range of 'Last 6 hours', and a search icon. A yellow box highlights the 'Stat' panel type selector in the top right. The main panel, also outlined in yellow, is titled 'Panel Title' and displays a large green number '12'. To the right is a sidebar with search options and panel configuration settings, including 'Panel options' (Title, Description, Transparent background, Panel links, Repeat options) and 'Value options' (Show, Calculation). At the bottom, the 'Query' tab is active, showing a data source of 'postgresql_grafana_test' and a query editor. A yellow box highlights the 'Code' button in the query editor. The query being entered is: `SELECT COUNT(*) as " " FROM gpmetrics.test('skew');`

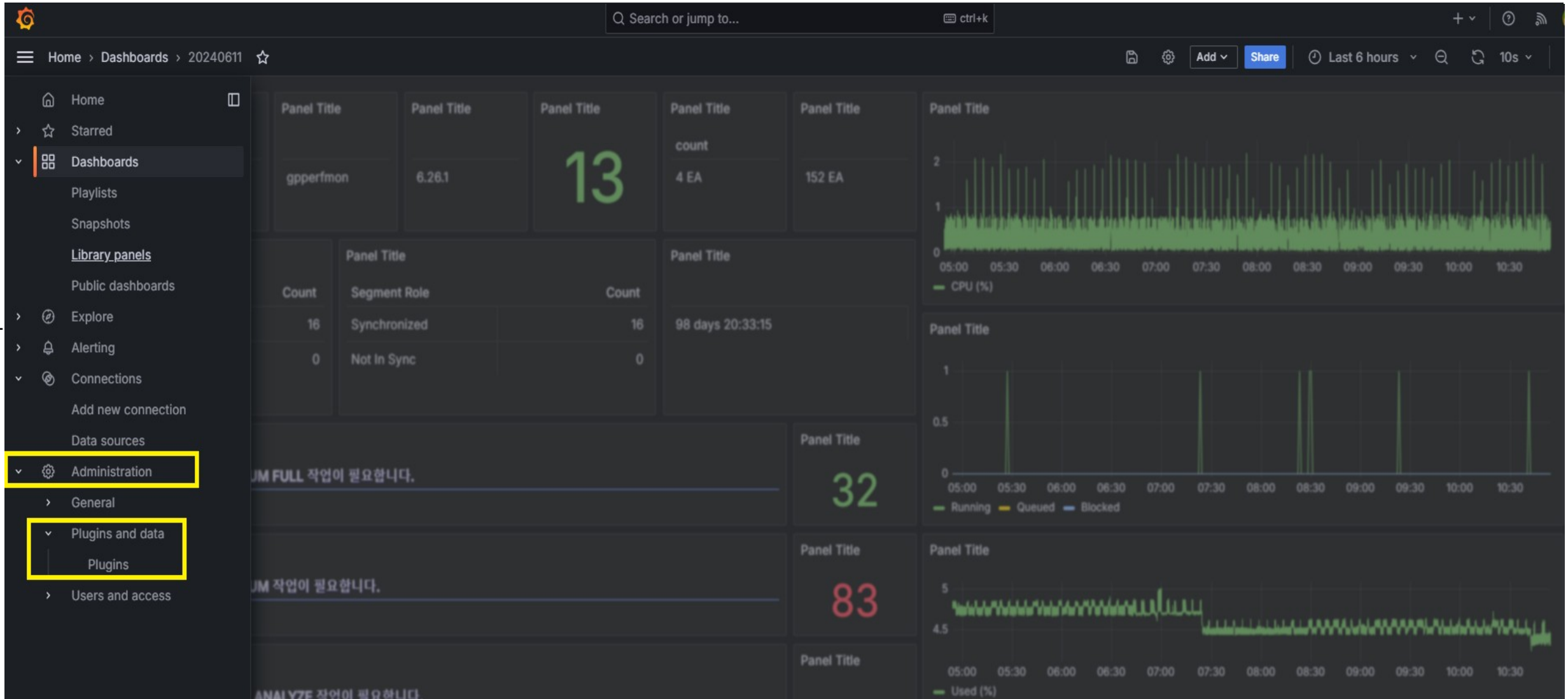
Grafana 구성 - 36

- 전체 Panel 구성 화면



Grafana 디자인 - 1

- 그래파나에서는 다양한 Plugin 기능이 존재하는데 그 중 Panels Plugin 을 통하여 간단한 디자인 작업 수행



Grafana 디자인 - 2

- Plugins 페이지 내 검색박스를 이용하여 간단한 디자인 작업에 필요한 Plugin Install 수행 -> “Boom Theme”, “Orchestra Cities Icon Stat Panel”, “HTML”

The screenshot shows the Grafana Plugins page for the 'Orchestra Cities Icon Stat Panel'. The page is dark-themed and features a navigation bar at the top with a hamburger menu, a breadcrumb trail (Home > Administration > Plugins and data > Plugins > Orchestra Cities Icon Stat Panel), and an 'Open menu' button. The main content area is divided into two sections: a header section and a description section. The header section includes a version number '12.4' next to the plugin name, which is highlighted with a yellow box. To the right of the name is a table with columns for Version, From, Downloads, Dependencies, and Signature. The 'Install' button is also highlighted with a yellow box. Below the header is a 'Website | License' link. The description section has tabs for 'Overview' and 'Version history'. The main content area displays the plugin's name 'Orchestra Cities - Icon Stat Panel' and a description: 'This plugin extends Grafana Stat panel with icons support. More info at <https://grafana.com/docs/grafana/latest/visualizations/stat-panel/>'. Below the description is a link to 'Icons supported are from FontAwesome'. The bottom section shows a preview of the plugin's output, consisting of four panels labeled 'Demo 1', 'Demo 2', 'Demo 3', and 'Demo 4'. Each panel displays a line chart with a green area under the line. The text 'prefix39.5% suffix' is overlaid on the charts in different colors and fonts. The left sidebar contains a search bar, a plus sign, and a list of icons for various Grafana features.

Home > Administration > Plugins and data > Plugins > Orchestra Cities Icon Stat Panel

Open menu

12.4 Orchestra Cities Icon Stat Panel

Website | License

Overview Version history

Orchestra Cities - Icon Stat Panel

This plugin extends Grafana Stat panel with icons support. More info at <https://grafana.com/docs/grafana/latest/visualizations/stat-panel/>

Icons supported are from [FontAwesome](#)

General / Demo ☆ 🔗

Demo 1

prefix39.5% suffix

Demo 2

availablespotnumber
prefix39.5% suffix

Demo 3

my title

Demo 4

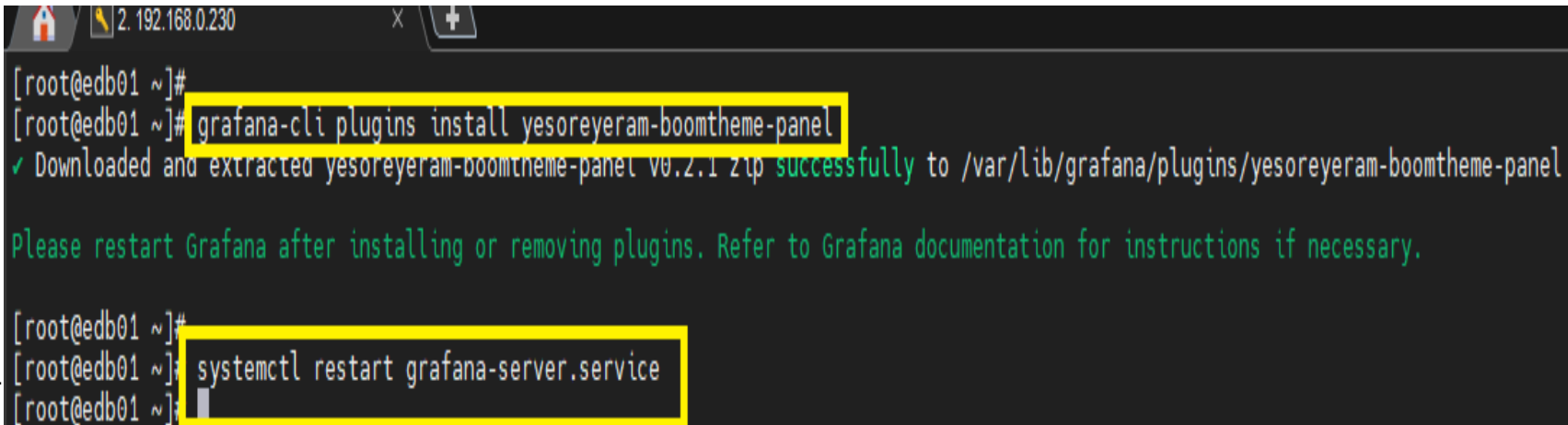
availablespotnumber

Grafana 디자인 - 3

- “Boom Theme” 의 경우 그라파나 서버 내 cli 설치

: <https://github.com/yesoreyeram/yesoreyeram-boomtheme-panel>

: grafana-cli plugins install yesoreyeram-boomtheme-panel -> Plugin Install 이후 그라파나 서비스 재기동

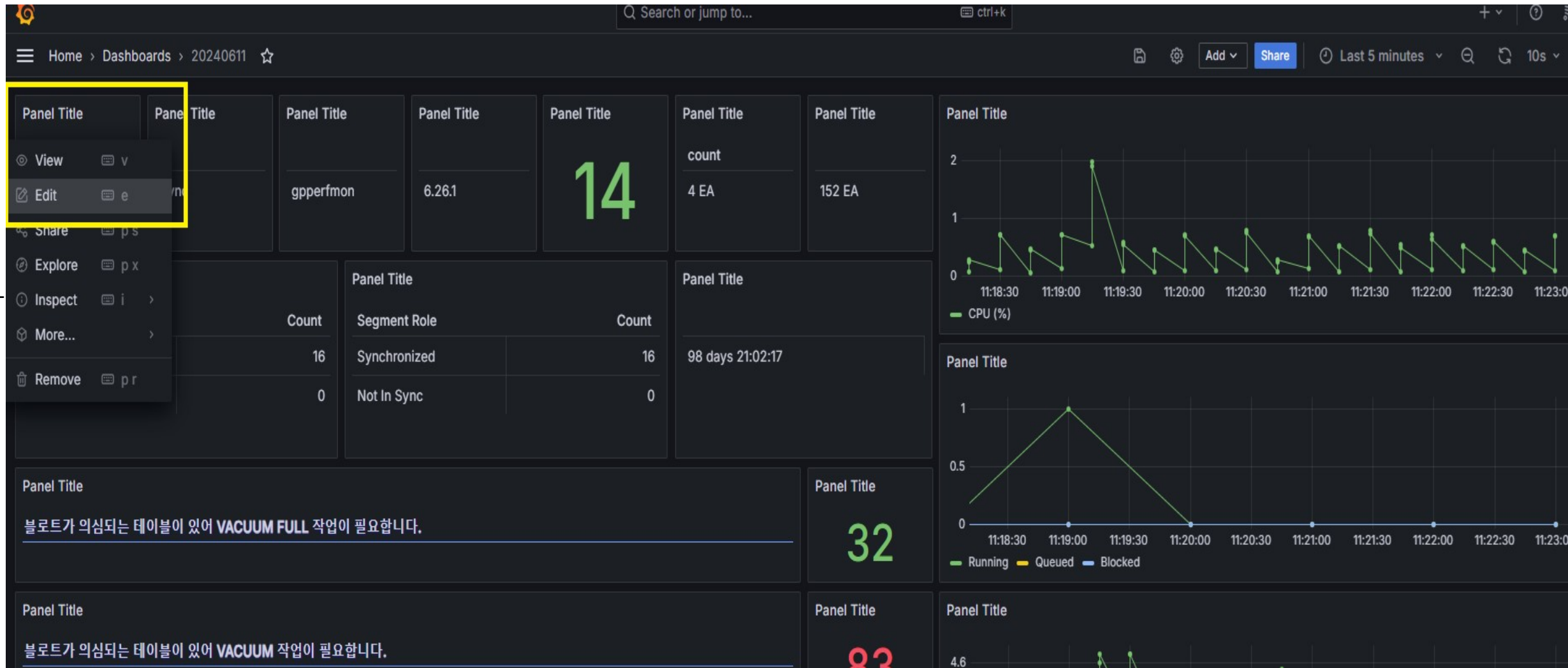


```
[root@edb01 ~]#  
[root@edb01 ~]# grafana-cli plugins install yesoreyeram-boomtheme-panel  
✓ Downloaded and extracted yesoreyeram-boomtheme-panel v0.2.1 zip successfully to /var/lib/grafana/plugins/yesoreyeram-boomtheme-panel  
  
Please restart Grafana after installing or removing plugins. Refer to Grafana documentation for instructions if necessary.  
  
[root@edb01 ~]#  
[root@edb01 ~]# systemctl restart grafana-server.service  
[root@edb01 ~]#
```

Grafana 디자인 - 4

- Panel 구성 완료 -> 디자인 작업 수행

: 수정하고자 하는 패널에다가 마우스 포인트를 가져다 대면 점자식의 Menu 아이콘이 보여지는데 클릭 후 “Edit” 선택



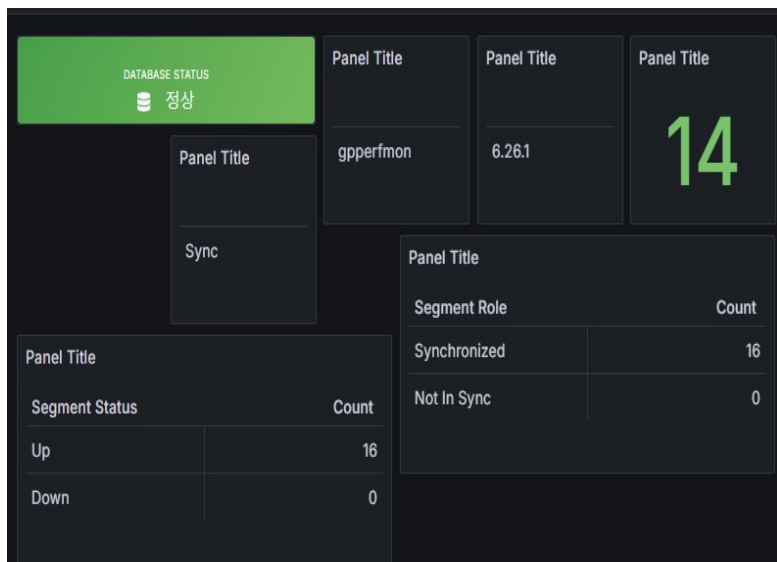
Grafana 디자인 - 5

- Edit 화면 내 우측 “Orchestra Cities Icon Stat Panel” 선택

: Title 공란 입력 -> Fields “All Fields” 선택

: 1) Orientation “Vertical” 선택 -> 2) Icon “database” 선택 -> 3) subtitle “DATABASE STATUS” 입력 -> 4) Color mode “Background” 선택 -> 5) Text alignment “Center” 선택

: 6) Text size value “15” 입력 -> 7) Color scheme “Red-Yellow-Green (By value)” 선택 -> 8) Value Mappings “정상 -> 정상 -> 초록색 / 비정상 -> 비정상 -> 빨간색” 입력



The configuration page for the 'Orchestra Cities Icon Stat Panel' shows various settings. The 'Title' field is empty. The 'Fields' section is set to 'All Fields'. The 'Color mode' is set to 'Background'. The 'Text alignment' is set to 'Center'. The 'Text size' is set to '15'. The 'Color scheme' is set to 'Red-Yellow-Green (by value)'. The 'Value mappings' section shows a mapping from '정상' to '정상' (green) and '비정상' to '비정상' (red).

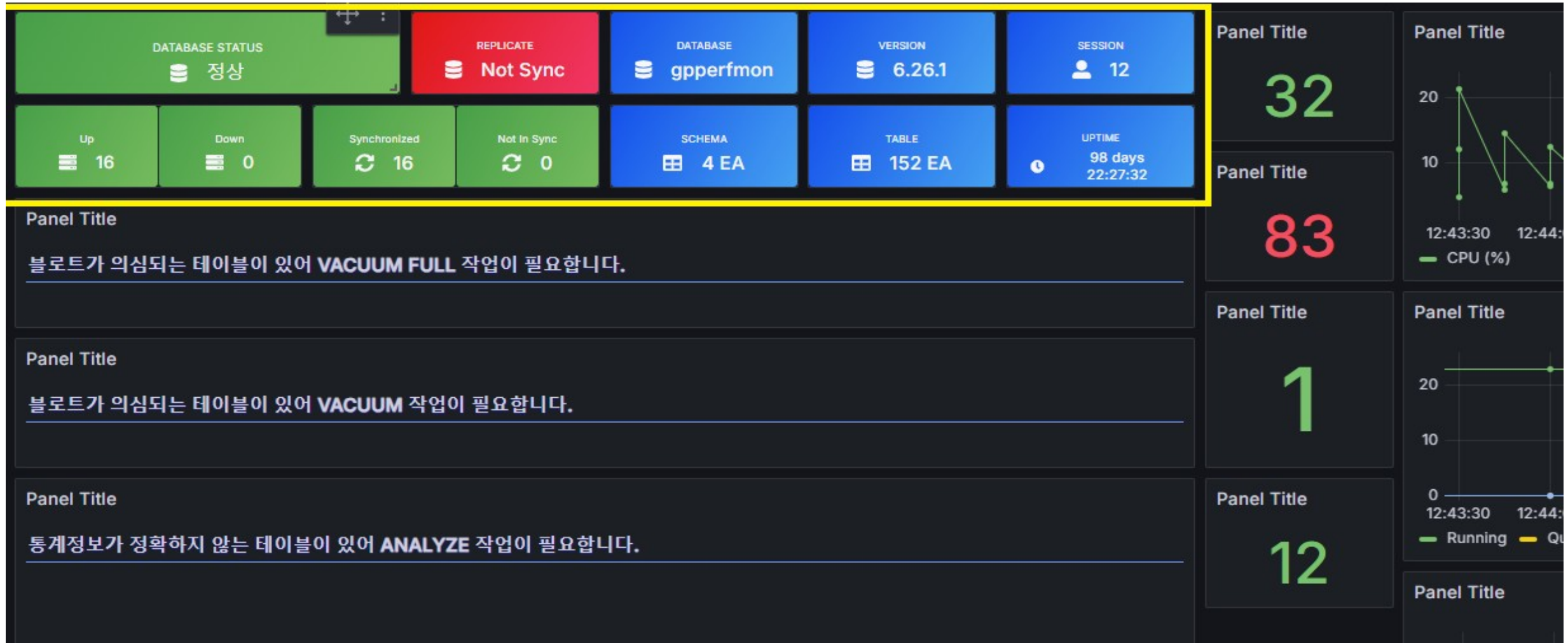
The 'Stat styles' configuration page shows settings for the panel's appearance. The 'Orientation' is set to 'Vertical'. The 'Icon' is set to 'database'. The 'Icon position' is set to 'content'. The 'subtitle' is set to 'DATABASE STATUS'. The 'Color mode' is set to 'Background'.

The configuration page shows the 'Color mode' set to 'Background' and the 'Text alignment' set to 'Center'. The 'Text size' is set to '15'.

The configuration page shows the 'Color scheme' set to 'Red-Yellow-Green (by value)'. The 'Value mappings' section shows a mapping from '정상' to '정상' (green) and '비정상' to '비정상' (red).

Grafana 디자인 - 6

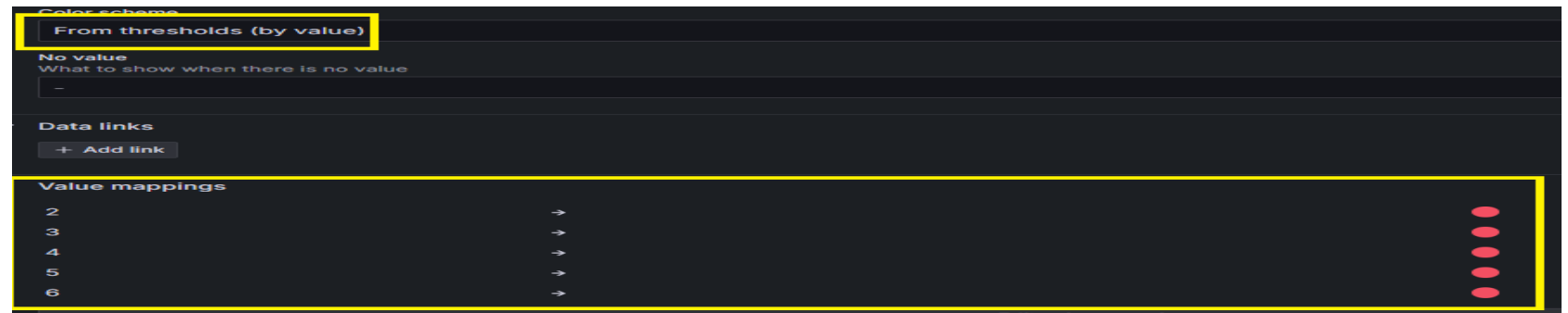
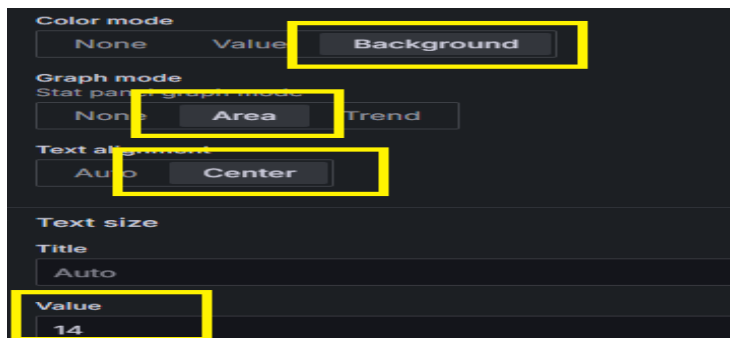
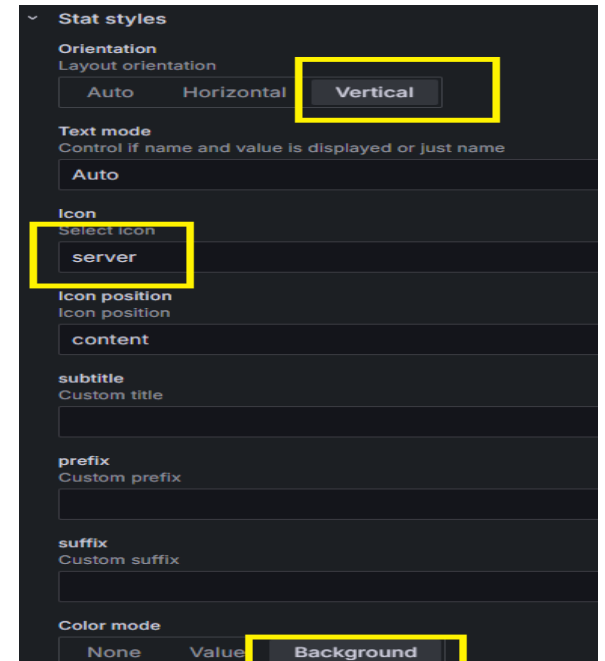
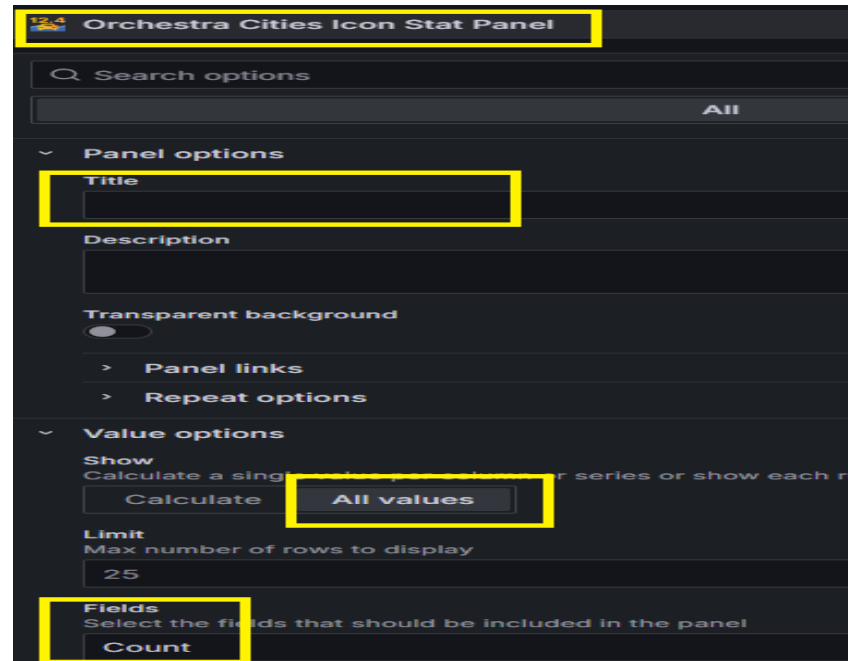
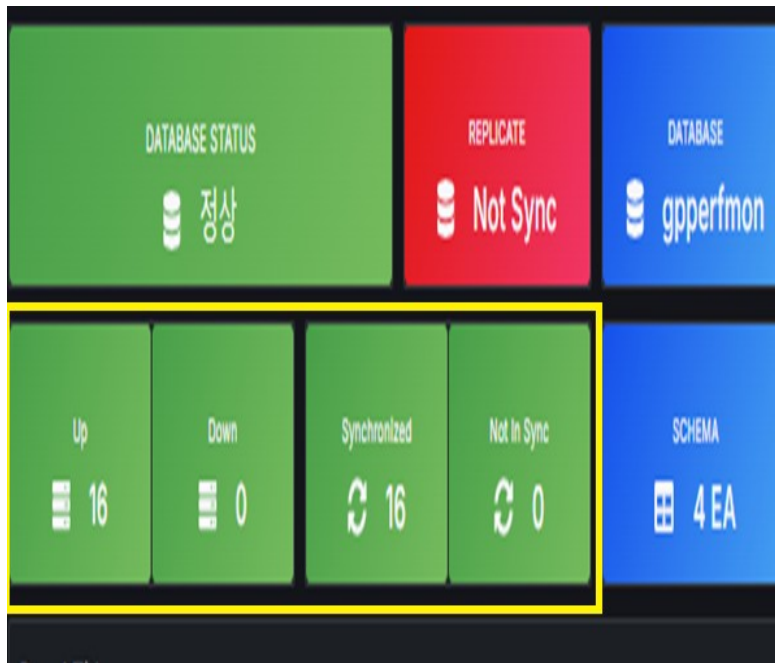
- DATABASE STATUS / REPLICATE / DATABASE / VERSION / SESSION / SCHEMA, TABLE, UPTIME 등 패널은 [디자인 -5] 항목과 동일하며 값만 원하는대로 변경해주면 됨
- : Value mappings 항목의 경우 지정된 쿼리 값에 따라 동적으로 동작하게끔 설정 Ex) REPLICATE 값이 “Sync” 라면 “Panel 색상이 초록색” / “Not Sync” 라면 “Panel 색상이 빨간색”
- : 설정항목들이 어려운데 아니므로 이것저것 만져보는 것이 좋음



Grafana 디자인 - 7

- INSTANCE STATUS (UP & DOWN) / INSTANCE SYNC STATUS (Synchronized & Not in Sync) 패널 수정

: 설정항목들이 어려운데 아니므로 이것저것 만져보는 것이 좋음



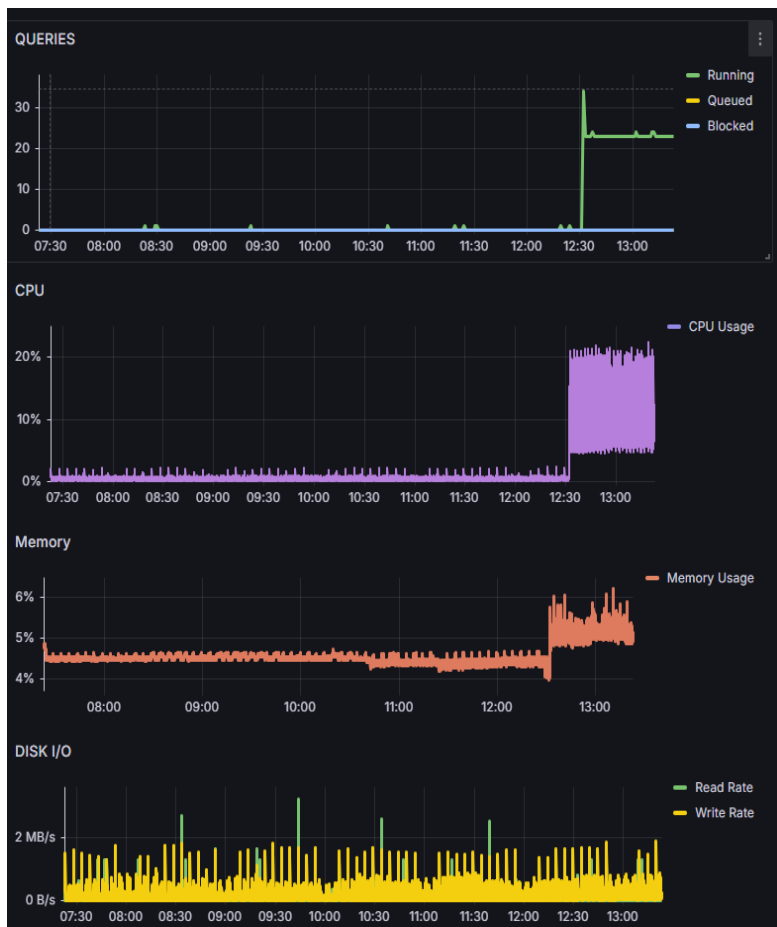
Grafana 디자인 - 8

- Chart Panel -> “Edit” 선택

: Title “CPU, Memory, DISK I/O” 등 차트 제목 입력 -> Transparent background “On” 선택 -> Legend Mode/Placement “List” 및 “Right” 선택

: 1) Timezone “Asia/Seoul” 선택 -> 2) Showbar “On” -> 3) Standard options Unit “Number / Percent (0-100) / bytes/sec(SI)” 등 데이터 단위 지정 -> 4) Decimals “0” 입력

: 설정항목들이 어려운데 아니므로 이것저것 만져보는 것이 좋음



Time series

Search options

All

Panel options

Title
QUERIES

Description

Transparent background
☒

Legend

Visibility
☒

Mode
List Table

Placement
Bottom Right

Width
Auto

Values
Select values or calculations to show in legend
Choose

Axis

Time zone
Asia/Seoul

Placement

Standard options

Unit
Number

Min
Leave empty to calculate based on all values
auto

Max
Leave empty to calculate based on all values
auto

Field min/max
Calculate min max per field
☐

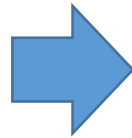
Decimals
0

Grafana 디자인 - 9

- COUNT PANEL 의 경우 [디자인 -5] 참조

: 세부 값은 아래 스크린샷 참조

: 설정항목들이 어려운데 아니므로 이것저것 만져보는 것이 좋음



The configuration interface for the 'Orchestra Cities Icon Stat Panel' includes the following sections:

- Panel options**: Title, Description, Transparent background (toggle), Panel links, Repeat options.
- Value options**: Show (Calculate a single value per column or series or show each row), Limit (Max number of rows to display), Fields (Select the fields that should be included in the panel).

The configuration interface for the 'Value' panel includes the following sections:

- Value**: Value (15).
- Standard options**: Unit (Choose), Min (Leave empty to calculate based on all values).

The configuration interface for 'Stat styles' includes the following sections:

- Orientation**: Layout orientation (Auto, Horizontal, Vertical).
- Text mode**: Control if name and value is displayed or just name (Auto).
- Icon**: Select icon (none).
- Icon position**: Icon position (content).
- subtitle**: Custom title (COUNT).
- prefix**: Custom prefix.
- suffix**: Custom suffix.
- Color mode**: None, Value, Background.

The configuration interfaces for 'Decimals' and 'Display name' include the following sections:

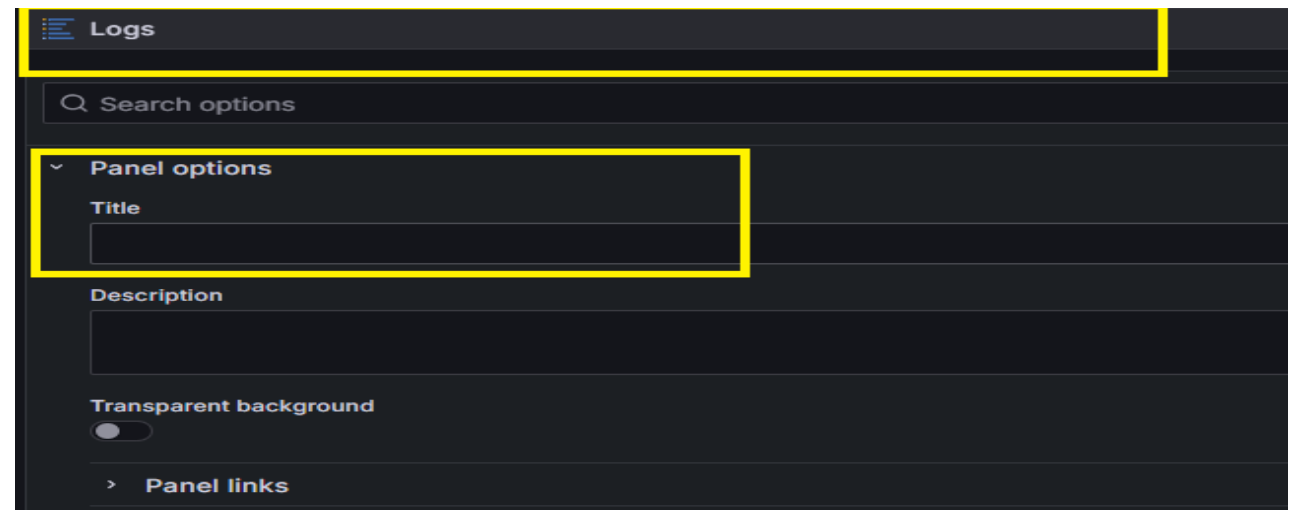
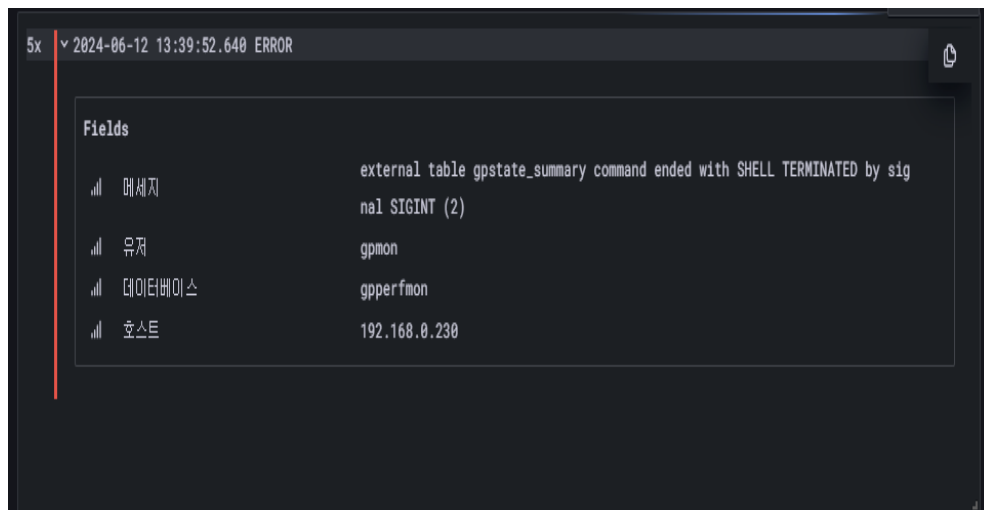
- Decimals**: auto.
- Display name**: Change the field or series name (none).
- Color scheme**: Blue-Purple (by value).

Grafana 디자인 - 10

- LOG PANEL 세부 내용

: 세부 값은 아래 스크린샷 참조

: 설정항목들이 어려운데 아니므로 이것저것 만져보는 것이 좋음



Grafana 디자인 - 11

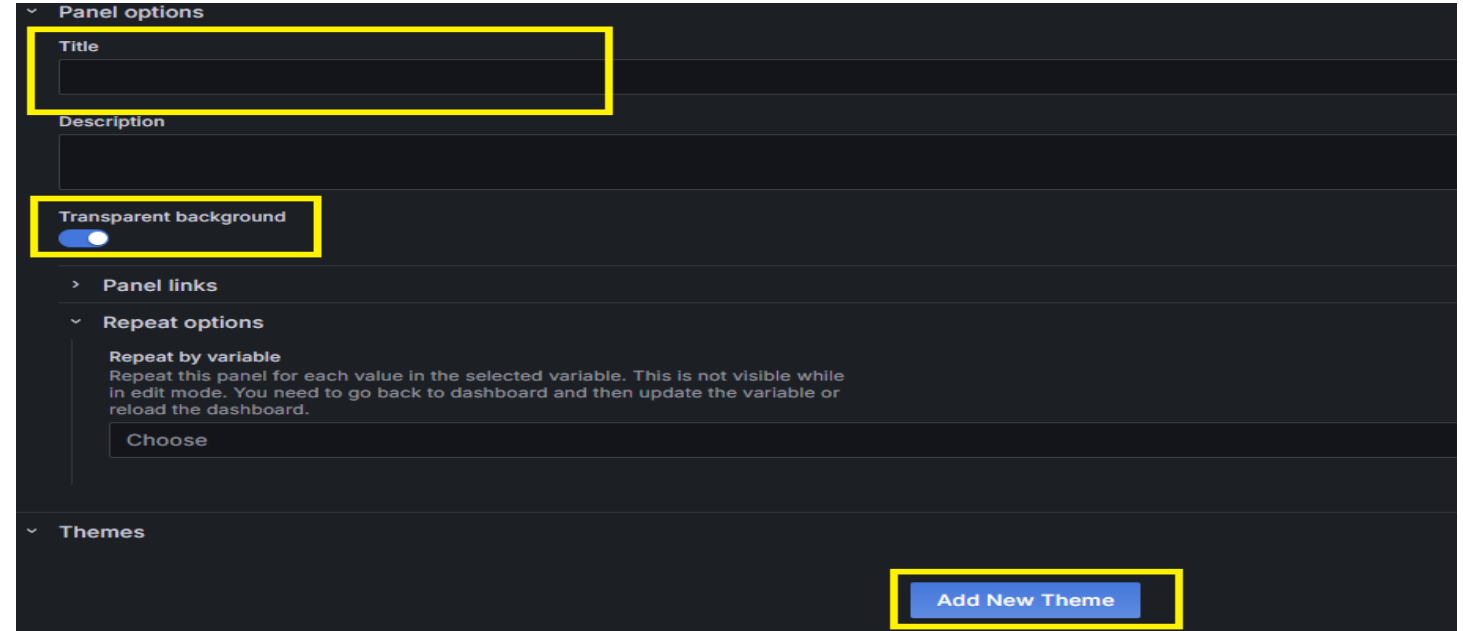
- 전체 Panel 구성 화면



Grafana 디자인 - 12

- 배경 및 공간 (Space) 디자인 작업을 위하여 기 설치한 “ Boom Theme” Plugin 이용

: Panel 추가 -> Boom Theme 선택 -> Panel 생성 후 Edit -> Title 공란 입력 및 Transparent background “ON” 선택 -> 하단 “ Add New Theme” 선택



Grafana 디자인 - 13

- “Add New Theme” 선택 시 -> “New Theme 1” 생성

: Write Icon (파란색) 선택

: 1) Theme 이름 입력 Ex) test_theme

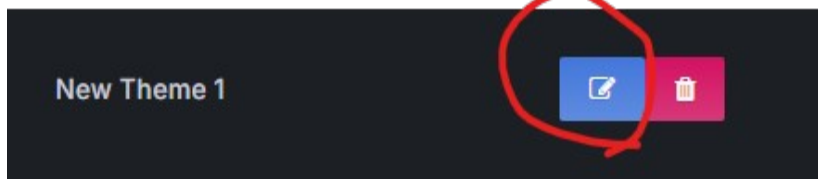
: 2) Background Image 값 지정 (다른 배경이미지 가져와서 써도됨)

-> Ex) https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQiJ4HpGL3RtbXCFh3B54U-zd9j8vW3tzFeikzBHOLMJ0cSBd9Zbx_MAO_yQ-VZ5BnJBKI&usqp=CAU

: 3) Additional CSS Sytle 값 지정 (배경 외 추가적으로 각 패널들에 대해서 컨트롤이 가능하므로 DOM 선택 후 CSS 코드 추가 삽입 가능)

Ex)

```
-> .panel-container {  
  border: none;  
  background-color:transparent;  
}
```



Dialog box titled "Edit test_theme" with the following fields:

- Theme Name: test_theme
- Base Theme: Default (selected), Dark, Light
- Background Image: https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQiJ4HpGL3RtbXCFh3B54U-zd9j8vW3tzFeikzBHOLMJ0cSBd9Zbx_MAO_yQ-VZ5BnJBKI&usqp=CAU
- External CSS URL: (empty)
- Additional CSS Style:

```
.panel-container {  
  border: none;  
  background-color:transparent;  
}
```

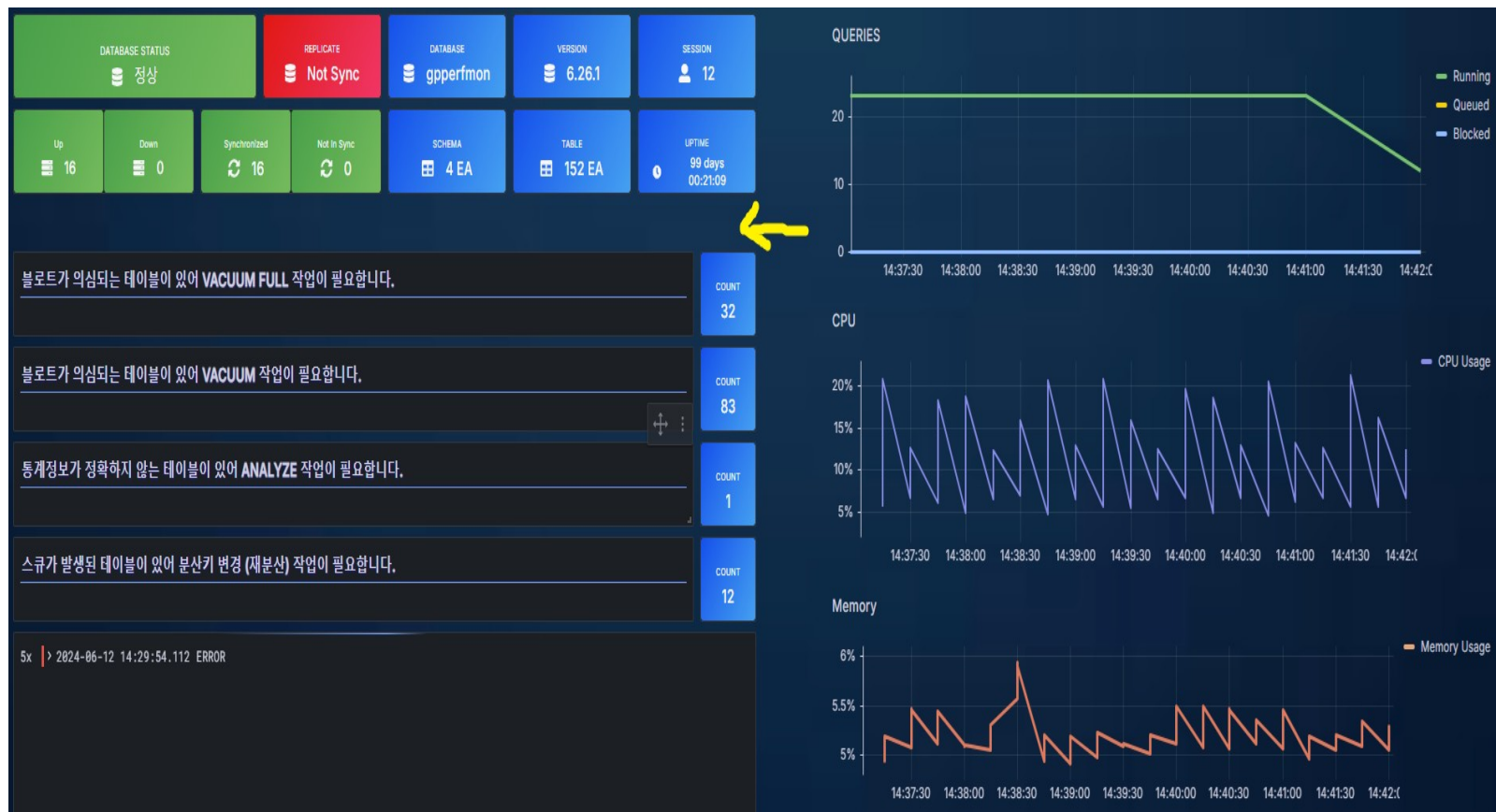
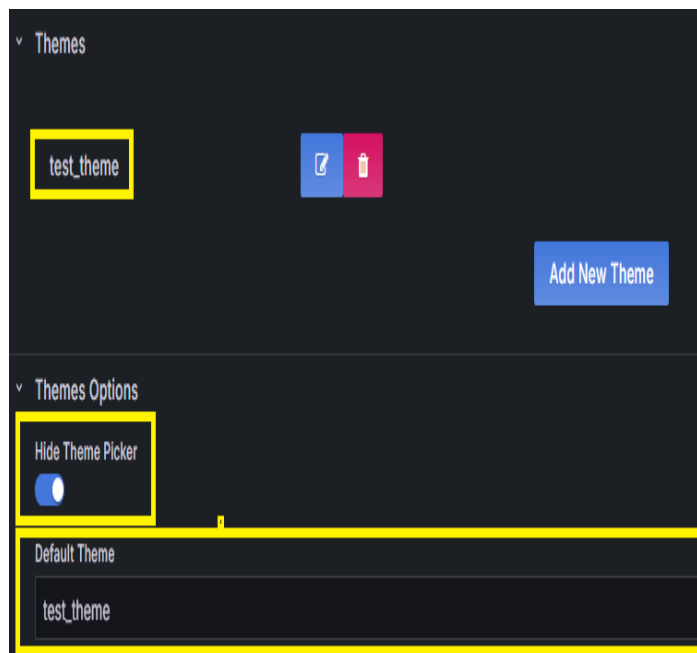
Buttons at the bottom: "Add Panel BG Color", "OK", "Cancel".

Grafana 디자인 - 14

- 이후 맨 하단 내 Themes Options -> “Hide Theme Picker” ON 선택

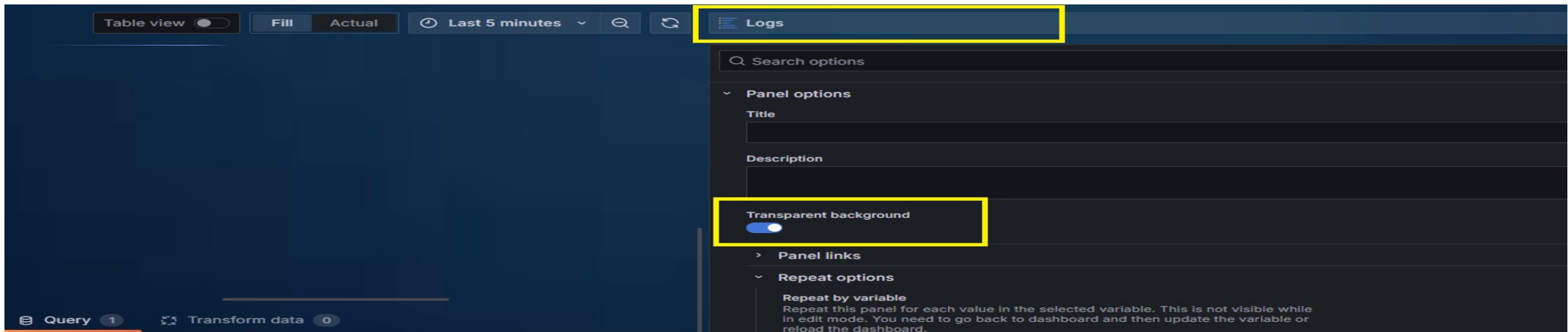
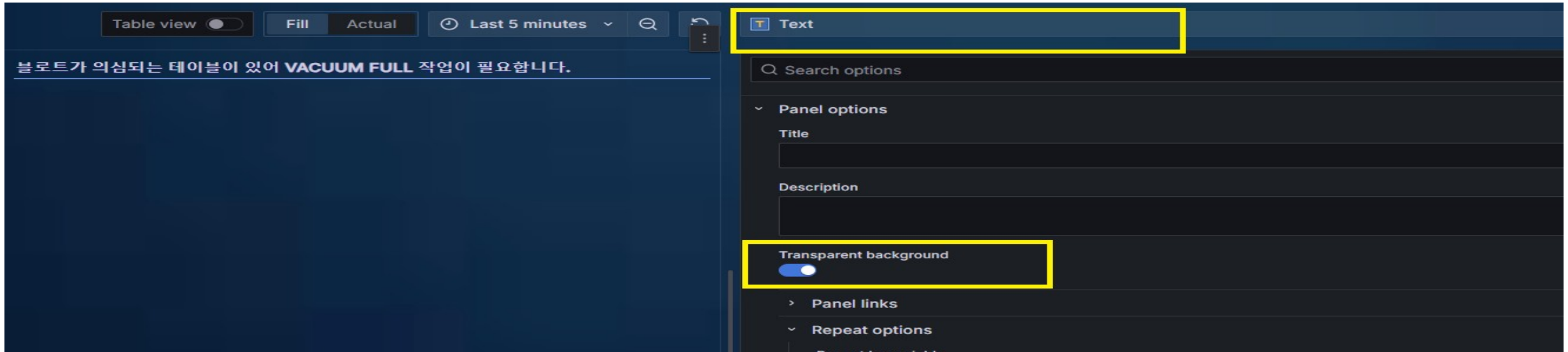
: Default Theme : “test_theme” 입력 (기본 테마를 설정하는 것으로 이전에 생성한 Theme 이름을 적어줘야 적용됨)

-> Apply 적용 시 배경 변경 -> 페이지 내 기 생성한 “ Boom Theme” Panel 이 보일텐데 이 Panel 의 크기 및 위치 조정하여 공간 디자인으로 활용
(아래 화면에서는 상단 Instance Status Panel 아래에 위치함)



Grafana 디자인 - 15

- 나머지 Text Panel 및 Log Panel 들의 Transparent background 의 값을 “ON” 적용



Grafana 디자인 - 16

- 대시보드 구성 완료



End of Document